

MIMIC-IV Data Analysis via medicalcoder

Supplement to JAMIA Open Manuscript

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1 Objective and Setup

This supplement provides reproducible validation and benchmarking of `medicalcoder::comorbidities()` using MIMIC-IV v3.1 data, including algorithmic agreement with established implementations and demonstration of longitudinal flagging behavior.

2 Materials and Methods

2.1 Data

We use the MIMIC-IV v3.1 Data (Johnson et al. 2024; Johnson et al. 2023; Goldberger et al. 2000) as the foundation for these examples.

The MIMIC-IV v3.1 dataset is a large, publicly available clinical database containing de-identified health data from patients admitted to intensive care units at Beth Israel Deaconess Medical Center. It includes detailed information on diagnoses, procedures, laboratory results, medications, and hospital encounters, with ICD-9-CM and ICD-10-CM/PCS codes available for billing and comorbidity research. MIMIC-IV is widely used for method development, validation, and reproducible research in clinical informatics and outcomes research.

The MIMIC-IV v3.1 data set can be obtained so long as you meet the requirements set by PhysioNet¹.

¹<https://physionet.org/content/mimiciv/3.1/>

Table 1: Comorbidity methods implemented in *medicalcoder* and sorted by comorbidity family and method in alphabetical order.

Comorbidity Method	Notes
Charlson	
charlson_beyrer2021	U.S. extension of Quan et al. (2005) by Beyrer et al. (2021)
charlson_cdmf2019	ICD-9 and ICD-10 defined in [glasheen2019]
charlson_deyo1992	ICD-9 codes defined in Table 1 of Quan et al. (2005)
charlson_mimicivcode	MIMIC-IV Charlson SQL from [mimic-code](https://github.com/MIT-LCP/mimic-code)
Elixhauser	
charlson_quan2005	ICD-9 and ICD-10 defined in Table 1 of Quan et al. (2005); index scoring as reported in Table 2 of Quan et al. (2011)
charlson_quan2011	ICD-9 and ICD-10 defined in Table 1 of Quan et al. (2005); index scoring as reported in Table 2 of Quan et al. (2011)
elixhauser_elixhauser1988	ICD-9 codes defined in Table 2 of Quan et al. (2005)
elixhauser_ahrq_web	ICD-9 codes defined in Table 2 of Quan et al. (2005)
elixhauser_quan2005	ICD-9 and ICD-10 defined in Table 2 of Quan et al. (2005)
elixhauser_ahrq2022	ICD-10 codes from AHRQ for fiscal year 2022
elixhauser_ahrq2023	ICD-10 codes from AHRQ for fiscal year 2023
elixhauser_ahrq2024	ICD-10 codes from AHRQ for fiscal year 2024
elixhauser_ahrq2025	ICD-10 codes from AHRQ for fiscal year 2025
PCCC	
elixhauser_ahrq2026	ICD-10 codes from AHRQ for fiscal year 2026
elixhauser_ahrq_icd10	Any ICD-10 code from all elixhauser_ahrqYYYY
pccc_v2.0	ICD-9 and ICD-10 diagnostic and procedure codes consistent with pccc::ccc()
pccc_v2.1	Extended set of pccc_v2.0 codes based on pccc::ccc() and supplements to Feudtner et al. (2014)
pccc_v3.0	ICD-9 and ICD-10 diagnostic and procedure codes consistent with SAS code from Children's Hospital Association
pccc_v3.1	Extended set of pccc_v3.0 codes based on the supplements to Feinstein et al. (2024)

2.2 Comorbidity Algorithms

Within the *medicalcoder* package are implementations for several variants of the Charlson, Elixhauser, and PCCC comorbidity algorithms, (Table 1) (Beyrer et al. 2021; Quan et al. 2005, 2011; Glasheen et al. 2019; Healthcare Research and (AHRQ) 2025; Feudtner et al. 2014; Feinstein et al. 2024; DeWitt et al. 2026; Feinstein et al. 2018; *Complex Chronic Conditions Version 3.0* 2024).

NOTE: The MIMIC-IV data consists of only adult patients. Although PCCC was designed for pediatric populations, adult MIMIC-IV data are used here solely to demonstrate algorithmic implementation.

2.3 Alternative Utilities

We compared the results from `medicalcoder::comorbidities()` to the Charlson comorbidities flagged by the MIMIC-IV code, and the R package *comorbidity* (Gasparini 2018). For Elixhauser we will compare results between `medicalcoder::comorbidities()` and `comorbidity::comorbidity()`. Lastly, we will compare PCCC version 2 results between `medicalcoder::comorbidities()` and the R package *pccc* (DeWitt et al. 2026) call `pccc::ccc()`.

Table 2: SHA256 Check sums for files in the MIMIC-IV v3.1 data.

sha256sum	file
a9584ed88e9ed664a2f66f86a5cf9fd175c8bb0af50e3b6115598b19e978384e	hosp/admissions.csv.gz
47665a41b2a3ad990d6f314062d5bd6d29d467b0fd402dd94662044ec8b073d2	hosp/diagnoses_icd.csv.gz
4c7507de7ecad8c5ba7647ea4d26018e88ff6672c5fc22ab59c2a5697d687cdd	hosp/patients.csv.gz
3271397d0517131f84399b698cbb0c2f435aaab0f873a3806e13b097661596f7	hosp/procedures_icd.csv.gz

3 Prepare MIMIC-IV Data For Analysis

We have stored the path to the MIMIC-IV data as a system variable on our machines as `MIMICIVDATA`. The following code is provide to help readers verify that they are using the same version of the MIMIC-IV v3.1 data as used in this document.

```
# verify that you are looking at version 3.1 of the MIMIC-IV data
stopifnot(
  identical(
    scan(
      file = file.path(Sys.getenv("MIMICIVDATA"), "CHANGELOG.txt"),
      what = "character", sep = "\n", quiet = TRUE)[1],
    "This is the change log for MIMIC-IV v3.1."
  )
)
```

We only need four of the MIMIC-IV hospital datasets: `admissions`, `patients`, `diagnoses_icd`, and `procedures_icd`.

We will do some light formatting before using `medicalcoder::comorbidities()` to flag comorbidities. Start by importing the `admissions`, `patients`, `diagnoses_icd`, and `procedures_icd`, datasets.

Table 2 gives the sha256 checksums for these files Each file checksum has been verified.

```
mimicivdata <-
  list.files(
    path = file.path(Sys.getenv("MIMICIVDATA"), "hosp"),
    pattern = "(admissions|patients|.+_icd)\\.csv\\.gz",
    full.names = TRUE
  )
names(mimicivdata) <- sub("\\.csv\\.gz$", "", basename(mimicivdata))

mimicivdata <- lapply(mimicivdata, data.table::fread)
```

We will build an additional data set for age (in years).

```
mimicivdata_admissions_age <-
  mimicivdata$admissions[, list(subject_id, hadm_id, admittance)]
mimicivdata_patients_age <-
  mimicivdata$patients[, list(subject_id, anchor_age, anchor_year)]

mimicivdata$ages <-
  merge(
    x = mimicivdata_admissions_age,
    y = mimicivdata_patients_age,
    all = FALSE,
    by = "subject_id"
  )
mimicivdata$ages[
  ,
  anchor_year := as.POSIXct(paste0(anchor_year, "-01-01"), format = "%Y-%m-%d")
]
mimicivdata$ages[
  ,
  age :=
    anchor_age +
    as.numeric(difftime(admittance, anchor_year, units = "days")) / 365.25
]
```

For the calls to `medicalcoder::comorbidities()` we can have all the ICD codes in one `data.frame` (`data.table` (Barrett et al. 2025) will be used here).

```
mimicivdata$icd <-
  data.table::rbindlist(
    list(dx = mimicivdata$diagnoses_icd, pr = mimicivdata$procedures_icd),
    idcol = "dx",
    use.names = TRUE,
    fill = TRUE
  )

# set the diagnoses indicator to an integer as expected for input to
# medicalcoder::comorbidities()
mimicivdata$icd[, dx := as.integer(dx == "dx")]
```

We put all the parts together into one `data.frame` (`data.table`)

```

mimicivDT <-
  merge(
    x = mimicivdata$patients[, .(subject_id)],
    y = mimicivdata$ages[, .(subject_id, hadm_id, age)],
    all = TRUE,
    by = c("subject_id")
  )

mimicivDT <-
  merge(
    x = mimicivdata$icd,
    y = mimicivDT,
    all = TRUE,
    by = c("subject_id", "hadm_id")
  )

mimicivDT <-
  merge(
    x = mimicivDT,
    y = mimicivdata$admissions[, .(subject_id, hadm_id, admittance)],
    all = TRUE,
    by = c("subject_id", "hadm_id")
  )

```

We need an encounter sequence variable, the `hadm_id` are not sequential.

NOTE: `admittance` can be used in the calls to `medicalcoder::comorbidities()` but it there is additional overhead and computational expense when sorting and joining by POSIXct variable. The `enc_seq` achieves the same utility with lower computational overhead.

```

data.table::setorder(mimicivDT, subject_id, admittance)
mimicivDT[, enc_seq := cumsum(!duplicated(hadm_id)), by = .(subject_id)]
data.table::setkey(mimicivDT, subject_id, enc_seq, hadm_id)

```

medicalcoder can use multiple id variables. Other utilities such as *comorbidity* and *pccc* can only use one id variable. As such we create `cid` to use in those examples.

```

mimicivDT[, cid := paste(subject_id, hadm_id, sep = "__")]

# Sanity checks:
# expect age to always be increasing as enc_seq increase
mimicivDT[

```

```

,
unique(.SD),
.SDcols = c("subject_id", "hadm_id", "admittime", "age", "enc_seq")
][
,
c(NA_real_, diff(age)),
by = .(subject_id)
][
,
all(V1 >= 0, na.rm = TRUE)
] |>
print() |>
stopifnot()
## [1] TRUE

```

Our final mimiciDT object:

```

str(mimiciDT)
## Classes 'data.table' and 'data.frame': 7365770 obs. of 11 variables:
## $ subject_id : int 10000032 10000032 10000032 10000032 10000032 10000032 10000032 10000032 10000032 10000032 ...
## $ hadm_id : int 22595853 22595853 22595853 22595853 22595853 22595853 22595853 22595853 22595853 22595853 ...
## $ dx : int 1 1 1 1 1 1 1 1 0 1 ...
## $ seq_num : int 1 2 3 4 5 6 7 8 1 1 ...
## $ icd_code : chr "5723" "78959" "5715" "07070" ...
## $ icd_version: int 9 9 9 9 9 9 9 9 9 9 ...
## $ chartdate : IDate, format: NA NA ...
## $ age : num 52.3 52.3 52.3 52.3 52.3 ...
## $ admittime : POSIXct, format: "2180-05-06 22:23:00" "2180-05-06 22:23:00" ...
## $ enc_seq : int 1 1 1 1 1 1 1 1 1 2 ...
## $ cid : chr "10000032_22595853" "10000032_22595853" "10000032_22595853" "10000032_22595853" ...
## - attr(*, ".internal.selfref")=<pointer: 0x105cbfbb0>
## - attr(*, "sorted")= chr [1:3] "subject_id" "enc_seq" "hadm_id"
summary(mimiciDT)
## subject_id hadm_id dx seq_num
## Min. :10000032 Min. :20000019 Min. :0.000 Min. : 1.000
## 1st Qu.:12510378 1st Qu.:22497017 1st Qu.:1.000 1st Qu.: 3.000
## Median :15003038 Median :25003884 Median :1.000 Median : 6.000
## Mean :15003453 Mean :25000556 Mean :0.881 Mean : 8.173
## 3rd Qu.:17503124 3rd Qu.:27500738 3rd Qu.:1.000 3rd Qu.:12.000
## Max. :19999987 Max. :29999935 Max. :1.000 Max. :41.000
## NAs :141175 NAs :141627 NAs :141627
## icd_code icd_version chartdate age

```

```
## Length :7365770 Min. : 9.000 Min. :2105-10-05 Min. : 18.00
## N.unique : 42768 1st Qu.: 9.000 1st Qu.:2133-12-25 1st Qu.: 53.19
## N.blank : 0 Median :10.000 Median :2153-10-14 Median : 65.96
## Min.nchar: 3 Mean : 9.532 Mean :2153-11-29 Mean : 64.08
## Max.nchar: 7 3rd Qu.:10.000 3rd Qu.:2173-11-11 3rd Qu.: 77.37
## NAs : 141627 Max. :10.000 Max. :2214-10-01 Max. :106.27
## NAs :141627 NAs :6506115 NAs :141175
## admittime enc_seq cid
## Min. :2105-10-04 17:26:00 Min. : 1.000 Length :7365770
## 1st Qu.:2135-07-28 17:21:00 1st Qu.: 1.000 N.unique : 687203
## Median :2155-05-12 02:46:00 Median : 2.000 N.blank : 0
## Mean :2155-07-19 16:07:22 Mean : 4.834 Min.nchar: 12
## 3rd Qu.:2175-08-11 02:01:00 3rd Qu.: 5.000 Max.nchar: 18
## Max. :2214-12-15 19:11:00 Max. :238.000
## NAs :141175
```

There are 7,365,770 total rows of data for 364,627 unique subjects and a total of 687,203 unique hospital encounters.

4 Investigating ICD Codes

The *medicalcoder* package provides utilities for investigating ICD codes. For the MIMIC-IV data, let's just check if the reported ICD codes are in the *medicalcoder* ICD data.

4.1 Utilities

There are several functions in the *medicalcoder* package for working with ICD codes:

- `get_icd_codes()`: returns a lookup table of ICD codes as a `data.frame`.
- `lookup_icd_codes()`: returns details on specific ICD codes.
- `is_icd()`: returns `TRUE` or `FALSE` for a vector of codes while considering ICD version, type, and billable status.
- `icd_compact_to_full()`: insert a decimal point into a string to be consistent with ICD-9 diagnostic, ICD-9 procedure, or ICD-10 diagnostic codes. (ICD-10 procedure codes do not have decimal places.) NOTE: this will not validate the return as a valid ICD code, just format the input string appropriately.

We encourage readers to review the vignette in *medicalcoder* for details on these utilities.

```
vignette(topic = "icd", package = "medicalcoder")
```

4.2 ICD-Codes in MIMIC-IV Data

Let's quickly verify that all the reported ICD codes in the MIMIC-IV data are known and ICD-9-CM, ICD-9-PCS, ICD-10-CM, or ICD-10-PCS codes.

```
unique_icdcodes <- mimicivDT[!is.na(icd_code), unique(.SD), .SDcols = c("icd_code", "icd_ver
```

The call `medicalcoder::is_icd()` will return TRUE or FALSE for a vector of strings if the input is a known ICD code based on version (`icdv = 9` or `icdv = 10`), diagnostic (`dx = 1`) or procedure (`dx = 0`) codes, and source.

```
unique_icdcodes[icd_version == 9 & dx == 0, iscms := medicalcoder::is_icd(x = icd_code, icdv = 9, dx = 0)]
unique_icdcodes[icd_version == 9 & dx == 1, iscms := medicalcoder::is_icd(x = icd_code, icdv = 9, dx = 1)]
unique_icdcodes[icd_version == 10 & dx == 0, iscms := medicalcoder::is_icd(x = icd_code, icdv = 10, dx = 0)]
unique_icdcodes[icd_version == 10 & dx == 1, iscms := medicalcoder::is_icd(x = icd_code, icdv = 10, dx = 1)]
```

There are some codes which appear to not be valid ICD codes.

```
unique_icdcodes[, .N, keyby = .(iscms)]
## Key: <iscms>
##      iscms      N
##   <lgcl> <int>
## 1: FALSE    34
## 2:  TRUE 43460
```

Let's investigate these codes. All of them are header codes, never assignable.

```
notcms <-
  merge(
    x = subset(medicalcoder::get_icd_codes(), src == "cms"),
    y = unique_icdcodes[!(iscms)],
    by.x = c('icdv', 'dx', 'code'),
    by.y = c('icd_version', 'dx', 'icd_code')
  )
head(notcms)
##   icdv dx code full_code src known_start known_end assignable_start
## 1  10  1  B96      B96 cms      2014      2026             NA
## 2  10  1 C722    C72.2 cms      2014      2026             NA
```

```
## 3 10 1 D40 D40 cms 2014 2026 NA
## 4 10 1 E11 E11 cms 2014 2026 NA
## 5 10 1 E66 E66 cms 2014 2026 NA
## 6 10 1 E760 E76.0 cms 2014 2026 NA
## assignable_end iscms
## 1 NA FALSE
## 2 NA FALSE
## 3 NA FALSE
## 4 NA FALSE
## 5 NA FALSE
## 6 NA FALSE
all(is.na(notcms[["assignable_start"]]))
## [1] TRUE
```

If `headerok = TRUE` in the call `medicalcoder::is_icd()`, then these header codes will be considered valid.

```
unique_icdcodes[!is.na(icd_code) & icd_version == 9 & dx == 0, iscmshdrok := medicalcoder::
unique_icdcodes[!is.na(icd_code) & icd_version == 9 & dx == 1, iscmshdrok := medicalcoder::
unique_icdcodes[!is.na(icd_code) & icd_version == 10 & dx == 0, iscmshdrok := medicalcoder::
unique_icdcodes[!is.na(icd_code) & icd_version == 10 & dx == 1, iscmshdrok := medicalcoder::

unique_icdcodes[, .N, keyby = .(iscms, iscmshdrok)]
## Key: <iscms, iscmshdrok>
##   iscms iscmshdrok      N
##   <lgcl>   <lgcl> <int>
## 1: FALSE      TRUE    34
## 2:  TRUE      TRUE  43460
```

Lastly, to get more detail on any of the codes you can use `medicalcoder::lookup_icd_codes()` or `medicalcoder::get_icd_codes()`.

```
medicalcoder::lookup_icd_codes(c("Z31", "S73109"))
##   input_code match_type icdv dx full_code code src known_start
## 1 S73109 compact_code 10 1 S73.109 S73109 cms 2014
## 2 Z31 full_code 10 1 Z31 Z31 cms 2014
## 3 Z31 full_code 10 1 Z31 Z31 ihacpa 2020
## 4 Z31 full_code 10 1 Z31 Z31 socialstyrelsen 1997
## 5 Z31 full_code 10 1 Z31 Z31 who 2008
##   known_end assignable_start assignable_end
## 1 2026 NA NA
```

```

## 2      2026      NA      NA
## 3      2026      NA      NA
## 4      2026      NA      NA
## 5      2021      NA      NA
subset(medicalcoder::get_icd_codes(with.description = TRUE), code == "Z31")
##      icdv dx full_code code      src known_start known_end
## 343173   10 1      Z31  Z31      cms      2014      2026
## 343174   10 1      Z31  Z31 socialstyrelsen 1997      2026
## 343175   10 1      Z31  Z31      ihacpa      2020      2026
## 343176   10 1      Z31  Z31      who      2008      2021
##      assignable_start assignable_end      desc
## 343173      NA      NA Encounter for procreative management
## 343174      NA      NA Fertilitetsfrämjande åtgärder
## 343175      NA      NA Procreative management
## 343176      NA      NA Procreative management
##      desc_start desc_end
## 343173      2014      2026
## 343174      1997      2026
## 343175      2020      2026
## 343176      2008      2021
subset(medicalcoder::get_icd_codes(with.description = TRUE), startsWith(code, "S73109"))
##      icdv dx full_code      code src known_start known_end assignable_start
## 247110   10 1  S73.109 S73109 cms      2014      2026      NA
## 247111   10 1  S73.109A S73109A cms      2014      2026      2014
## 247112   10 1  S73.109D S73109D cms      2014      2026      2014
## 247113   10 1  S73.109S S73109S cms      2014      2026      2014
##      assignable_end
## 247110      NA
## 247111      2026
## 247112      2026
## 247113      2026
##      desc desc_start
## 247110      Unspecified sprain of unspecified hip 2014
## 247111      Unspecified sprain of unspecified hip, initial encounter 2014
## 247112      Unspecified sprain of unspecified hip, subsequent encounter 2014
## 247113      Unspecified sprain of unspecified hip, sequela 2014
##      desc_end
## 247110      2026
## 247111      2026
## 247112      2026
## 247113      2026

```

5 Flagging Encounter Level Comorbidities

In the following we will outline how to apply the comorbidities methods to the `mimicivDT` dataset, focusing on encounter level flagging of comorbidities. We start by showing how to apply the Charlson (Quan et al. 2005), Elixhauser (Quan et al. 2005), and PCCC v2 (Feudtner et al. 2014) methods to the `mimicivDT` dataset using the `medicalcoder::comorbidities()` call. Following that we will show how to do the same work using SQL code provided by MIMIC-code (Johnson et al. 2018), the R package *comorbidity* (Gasparini 2018), and the R package *pccc* (DeWitt et al. 2026). Lastly, we compare and contrast the differences in the flagged comorbidities between the methods.

5.1 `medicalcoder::comorbidities()`

5.1.1 Charlson Comorbidities

There are several variants of Charlson like comorbidity methods implemented in *medicalcoder*. One way to see the current list of implemented methods is to look at the names of the lookup table mapping ICD codes to conditions:

```
grep(
  pattern = "^charlson",
  x = names(medicalcoder::get_charlson_codes()),
  value = TRUE
)
## [1] "charlson_beyrer2021"      "charlson_cdmf2019"
## [3] "charlson_deyo1992"       "charlson_ludvigsson2021"
## [5] "charlson_mimicivcode"    "charlson_quan2005"
## [7] "charlson_sundararajan2004" "charlson_quan2011"
```

The `charlson_quan2011` method will be applied to the `mimicivDT` dataset and the results will be compared to the results from the `comorbidity::comorbidity()` method.

```
medicalcoder_charlson_quan2011 <-
  medicalcoder::comorbidities(
    data      = mimicivDT, # object inheriting data.frame
    icd.codes = "icd_code", # character string name of icd codes in data
    id.vars   = c("subject_id", "hadm_id"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    poa       = 1L, # consider all codes present on admission
    primarydx = 0L, # consider all diagnosis codes secondary diagnoses.
```

```

    method      = "charlson_quan2011",
    flag.method = "current" # default
  )
data.table::setkey(medicalcoder_charlson_quan2011, subject_id, hadm_id)

```

The SQL code for applying Charlson comorbidities to the MIMIC-IV data provided by [MIT-LCP](#) has a slightly different set of ICD-10 codes mapping to conditions. Quan et al. (2005, 2011) is based on ICD-10 codes, not ICD-10-CM. The MIMIC-IV SQL code includes the C4A category which is a ICD-10-CM specific category (see [this github issue](#)). To be consistent with the expected set of ICD codes identified in the Quan papers and the MIMIC-IV SQL code, *medicalcoder* provides distinct methods.

```

medicalcoder_charlson_mimiciv <-
  medicalcoder::comorbidities(
    data      = mimicivDT, # object inheriting data.frame
    icd.codes = "icd_code", # character string name of icd codes in data
    id.vars   = c("subject_id", "hadm_id"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    age.var   = "age",
    poa       = 1L,      # consider all codes present on admission
    primarydx = 0L,      # consider all diagnosis codes secondary diagnoses.
    method    = "charlson_mimicivcode",
    flag.method = "current" # default
  )
data.table::setkey(medicalcoder_charlson_mimiciv, subject_id, hadm_id)

```

5.1.2 Elixhauser Comorbidities

As with the Charlson comorbidities there are multiple variants of Elixhauser like comorbidity methods implemented in *medicalcoder*.

```

grep(
  pattern = "^elixhauser",
  x = names(medicalcoder::get_elixhauser_codes()),
  value = TRUE
)
## [1] "elixhauser_ahrq_web"          "elixhauser_elixhauser1988"
## [3] "elixhauser_quan2005"         "elixhauser_ahrq2022"
## [5] "elixhauser_ahrq2023"         "elixhauser_ahrq2024"

```

```
## [7] "elixhauser_ahrq2025"      "elixhauser_ahrq2026"  
## [9] "elixhauser_ahrq_icd10"
```

To apply the `elixhauser_quan2005` version, which is comparable to the R package *comorbidity*, do the following:

```
medicalcoder_elixhauser_quan2005 <-  
  medicalcoder::comorbidities(  
    data      = mimicivDT,  
    icd.codes = "icd_code",  
    id.vars   = c("subject_id", "hadm_id"),  
    icdv.var  = "icd_version",  
    dx.var    = "dx",  
    poa       = 1L,  
    primarydx = 0L,  
    method    = "elixhauser_quan2005"  
  )  
data.table::setkey(medicalcoder_elixhauser_quan2005, subject_id, hadm_id)
```

5.1.3 PCCC v2.0

PCCC v2.0 has been implemented to be consistent with the R package *pccc*. The v2.1 variants extends the ICD to conditions mapping while using the same underlying algorithm. v3.0 and v3.1 use a different set of ICD codes and algorithm for flagging conditions. For this work we focus on v2.0 so we can compare it to the results from the R package *pccc*.

```
medicalcoder_pccc_v2.0 <-  
  medicalcoder::comorbidities(  
    data      = mimicivDT,  
    icd.codes = "icd_code",  
    id.vars   = c("subject_id", "hadm_id"),  
    icdv.var  = "icd_version",  
    dx.var    = "dx",  
    poa       = 1L,  
    method    = "pccc_v2.0"  
  )  
data.table::setkey(medicalcoder_pccc_v2.0, subject_id, hadm_id)
```

5.2 MIMIC-IV Code

We use the SQL code for applying the Charlson comorbidity algorithm to the MIMIC-IV data provided by the MIT Laboratory for Computational Physiology (MIT_LCP) on [GitHub](#). We stored the path to a local copy of the repository in the system environmental variable MIMICIVCODE. The specific version of the SQL code being used is from commit [278df75ec30991ff3a6f5ceb6d2221635a085e9f](#)

```
mimiciv_charlson_sql <-
  file.path(Sys.getenv("MIMICIVCODE"), "concepts", "comorbidity", "charlson.sql")

# checksum
stopifnot(
  substr(
    system(
      sprintf("sha256sum %s", mimiciv_charlson_sql),
      intern = TRUE
    ),
    start = 1L,
    stop = 64L
  ) == "5b797b673ace4dc6f686848d24371382f49ac61fcc39101f09c3b42969d801"
)
```

The SQL code is expected to run on Google Big Query. We make a few small modifications to the code so we can evaluate the SQL locally via RSQLite.

```
# scan in the sql code
mimiciv_charlson_sql <-
  scan(
    file = mimiciv_charlson_sql,
    what = "character",
    sep = "\n",
    quiet = TRUE
  )

# modify the query to work in SQLite
# replace Google Big Query table names with table names to be using in the local
# RSQLite in memory database.
mimiciv_charlson_sql <-
  gsub(
    pattern = "physionet-data.mimiciv_hosp.admissions",
    replacement = "admissions",
```

```

    x = mimiciv_charlson_sql,
    fixed = TRUE
  )

mimiciv_charlson_sql <-
  gsub(
    pattern = "physionet-data.mimiciv_hosp.diagnoses_icd",
    replacement = "diagnoses",
    x = mimiciv_charlson_sql,
    fixed = TRUE
  )

mimiciv_charlson_sql <-
  gsub(
    pattern = "physionet-data.mimiciv_derived.age",
    replacement = "ages",
    x = mimiciv_charlson_sql,
    fixed = TRUE
  )

# Replace the Google Big Query SQL function GREATEST with MAX
mimiciv_charlson_sql <-
  gsub(
    pattern = "GREATEST",
    replacement = "MAX",
    x = mimiciv_charlson_sql,
    fixed = TRUE
  )

mimiciv_charlson_sql <- paste(mimiciv_charlson_sql, collapse = "\n")

```

We set up an SQL database in memory and evaluate the code.

```

con <- odbc::dbConnect(drv = RSQLite::SQLite(), dbname = ":memory:")

# add data to the data base
odbc::dbWriteTable(conn = con, name = "diagnoses", value = mimicivdata$diagnoses)
odbc::dbWriteTable(conn = con, name = "admissions", value = mimicivdata$admissions)
odbc::dbWriteTable(conn = con, name = "patients", value = mimicivdata$patients)
odbc::dbWriteTable(conn = con, name = "ages", value = mimicivdata$ages)

# Run the query and record the time required to do so

```

```
mimiciv_charlson <- odbc::dbGetQuery(con, mimiciv_charlson_sql)
# close DB connection
odbc::dbDisconnect(conn = con)
data.table::setDT(mimiciv_charlson)
data.table::setkey(mimiciv_charlson, subject_id, hadm_id)
```

5.3 comorbidity::comorbidity()

The R package *comorbidity* (Gasparini 2018) is a widely used and referenced tool for applying the Charlson and Elixhauser comorbidities. One consideration for using `comorbidity::comorbidity()` is that ICD-9 or ICD-10 codes are used independently. There are 30921 subjects with both ICD-9 and ICD-10 codes in their records, and 11 encounters with both ICD versions.

To flag the conditions appropriately we must split the data by ICD version, apply `comorbidity::comorbidity()` to each split, and then aggregate the results. Also, `comorbidity::comorbidity()` only takes one column for the id and thus we need to use `cid` for the id instead of the `c("subject_id", "hadm_id")` used in the calls to `medicalcoder::comorbidities()`. In the following work we will copy the data built for the encounter level flagging and extend for cumulative flagging as well.

The following is done using *comorbidity* version 1.1.0.

5.3.1 Charlson

```
comorbidity_charlson_icd9 <-
  comorbidity::comorbidity(
    x      = subset(mimicivDT, icd_version == 9 & dx == 1L),
    id     = "cid",
    code   = "icd_code",
    map    = "charlson_icd9_quan",
    assign0 = TRUE # set less severe comorbidities flags to 0 when more severe
                  # comorbidities is also flagged
  )

comorbidity_charlson_icd10 <-
  comorbidity::comorbidity(
    x      = subset(mimicivDT, icd_version == 10 & dx == 1L),
    id     = "cid",
    code   = "icd_code",
```

```

    map      = "charlson_icd10_quan",
    assign0 = TRUE
  )

comorbidity_charlson <-
  rbind(
    comorbidity_charlson_icd9,
    comorbidity_charlson_icd10
  )

# aggregate between the icd versions
comorbidity_charlson <-
  aggregate(. ~ cid, data = comorbidity_charlson, FUN = max)

# apply the scoring, to use comorbidity::score the object needs to have certain
# attributes which were lost when aggregating over the icd versions. Replace
# the attributes
attributes(comorbidity_charlson)[c("class", "variable.labels", "map")] <-
  attributes(comorbidity_charlson_icd9)[c("class", "variable.labels", "map")]

comorbidity_charlson[["score"]] <-
  comorbidity::score(
    x = comorbidity_charlson,
    weights = "quan",
    assign0 = TRUE
  )

# expand the id columns
data.table::setDT(comorbidity_charlson)
comorbidity_charlson[
  ,
  c("subject_id", "hadm_id") :=
    lapply(data.table::tstrsplit(cid, "__"), as.integer)
]
data.table::set(comorbidity_charlson, j = "cid", value = NULL)
data.table::setkey(comorbidity_charlson, subject_id, hadm_id)

```

5.3.2 Elixhauser

Compared to the work done for flagging Charlson comorbidities, Elixhauser flagging requires changing the `map` argument in the calls to `comorbidity::comorbidity()` and there is no

scoring to be done.

```
comorbidity_elixhauser_icd9 <-
  comorbidity::comorbidity(
    x      = subset(mimicivDT, icd_version == 9 & dx == 1L),
    id     = "cid",
    code   = "icd_code",
    map    = "elixhauser_icd9_quan",
    assign0 = TRUE
  )

comorbidity_elixhauser_icd10 <-
  comorbidity::comorbidity(
    x      = subset(mimicivDT, icd_version == 10 & dx == 1L),
    id     = "cid",
    code   = "icd_code",
    map    = "elixhauser_icd10_quan",
    assign0 = TRUE
  )

comorbidity_elixhauser <-
  rbind(
    comorbidity_elixhauser_icd9,
    comorbidity_elixhauser_icd10
  )

# aggregate between the icd versions
comorbidity_elixhauser <-
  aggregate(. ~ cid, data = comorbidity_elixhauser, FUN = max)

# expand the id columns
data.table::setDT(comorbidity_elixhauser)
comorbidity_elixhauser[
  ,
  c("subject_id", "hadm_id") :=
    lapply(data.table::tstrsplit(cid, "__"), as.integer)
]
data.table::set(comorbidity_elixhauser, j = "cid", value = NULL)
data.table::setkey(comorbidity_elixhauser, subject_id, hadm_id)
```

5.4 pccc::ccc()

The R package *pccc* (DeWitt et al. 2026) implements PCCC v2 (Feudtner et al. 2014). As with `comorbidity::comorbidity()` and many other implementations of comorbidities, only one version of ICD is assessed at a time.

The expected input structure of the data for `pccc::ccc()` is a wide format, one row per encounter, one column for each ICD code. The following uses *pccc* version 1.0.7.

```
mimicivDT_for_pccc <-
  split(mimicivDT, by = "icd_version") |>
  lapply(
    data.table::dcast,
    formula = cid ~ ifelse(dx == 1, "DX", "PR") + seq_num,
    value.var = "icd_code"
  )

pccc_pcccv2.0 <-
  rbind(
    pccc::ccc(
      data = mimicivDT_for_pccc[["9"]],
      id = cid,
      dx_cols = grep("^DX", names(mimicivDT_for_pccc[["9"]]), value = TRUE),
      pc_cols = grep("^PR", names(mimicivDT_for_pccc[["9"]]), value = TRUE),
      icdv = 9
    ),
    ,
    pccc::ccc(
      data = mimicivDT_for_pccc[["10"]],
      id = cid,
      dx_cols = grep("^DX", names(mimicivDT_for_pccc[["10"]]), value = TRUE),
      pc_cols = grep("^PR", names(mimicivDT_for_pccc[["10"]]), value = TRUE),
      icdv = 10
    )
  )

# aggregate over ICD version
pccc_pcccv2.0 <- aggregate(. ~ cid, data = pccc_pcccv2.0, FUN = max)

# expand the id columns
data.table::setDT(pccc_pcccv2.0)
pccc_pcccv2.0[
  ,
```

```

  c("subject_id", "hadm_id") :=
    lapply(data.table::tstrsplit(cid, "__"), as.integer)
]
data.table::set(pccc_pcccv2.0, j = "cid", value = NULL)
data.table::setkey(pccc_pcccv2.0, subject_id, hadm_id)

```

6 Comparing Encounter-level Flags Between Utilities

6.1 Charlson Comorbidities

6.1.1 `medicalcoder::comorbidities()` vs `comorbidity::comorbidity()`

The first thing to not is that the number of row returned by the two implementations are different.

```

nrow(medicalcoder_charlson_quan2011)
## [1] 687203
nrow(comorbidity_charlson)
## [1] 545497

```

The reason for this is that there are encounters with no diagnostic ICD codes. `medicalcoder::comorbidities()` returns a row for these encounters and `comorbidity::comorbidity()` does not. For these encounters, `medicalcoder::comorbidities()` returns a row with zero comorbidities.

```

# subject/hadm_id not in comorbidity::comorbidity() results all have no
# diagnostic codes.
mimiciDT[!comorbidity_charlson][, any(dx == 1 & !is.na(icd_code))]
## [1] TRUE
medicalcoder_charlson_quan2011[!comorbidity_charlson][, all(cmrb_flag == 0L)]
## [1] TRUE

```

We build the `mdcr_v_cmrb` object to make comparing the results between the `medicalcoder::comorbidities()` call and the `comorbidity::comorbidity()` call easier. We will only include the `subject_id/hadm_id` reported on by both.

```

mdcr_v_cmrb <-
  merge(
    x = medicalcoder_charlson_quan2011,

```

```

y = comorbidity_charlson,
all = FALSE,
by = c("subject_id", "hadm_id"),
suffixes = c("_mdcr", "_cmrb")
)

```

The relevant columns to compare are as follows:

```

mvc_columns <-
data.table::fread(text = "
  medicalcoder | comorbidity
  aidshiv      | aids
  cebvd        | cevd
  chf_mdcr     | chf_cmrb
  hp_mdcr      | hp_cmrb
  copd         | cpd
  dem          | dementia
  dm           | diab
  dmc          | diabwc
  mal          | canc
  mi_mdcr      | mi_cmrb
  mld_mdcr     | mld_cmrb
  msld_mdcr    | msld_cmrb
  mst          | metacanc
  pud_mdcr     | pud_cmrb
  pvd_mdcr     | pvd_cmrb
  rhd          | rheumd
  rnd          | rend
  cci          | score
")

```

If the columns are identical, we remove them from the `mdcr_v_cmrb` object. This test includes the class of the column. `medicalcoder::comorbidities()` returns integer values as does `comorbidity::comorbidity()` save the `score` columns which is numeric.

```

data.table::set(
  x = mdcr_v_cmrb,
  j = "score",
  value = as.integer(mdcr_v_cmrb[["score"]])
)

for (i in seq_len(nrow(mvc_columns))) {

```

```

x <- mvc_columns[["medicalcoder"]][i]
y <- mvc_columns[["comorbidity"]][i]
z <- identical(mdcv_v_cmr[["x"]], mdcv_v_cmr[["y"]])
if (z) {
  message(sprintf("`%s` and `%s` are identical.", x, y))
  data.table::set(mdcv_v_cmr, j = x, value = NULL)
  data.table::set(mdcv_v_cmr, j = y, value = NULL)
} else {
  stop(sprintf("`%s` and `%s` are not identical.", x, y))
}
}

## `aidshiv` and `aids` are identical.
## `cebud` and `cevd` are identical.
## `chf_mdcv` and `chf_cmr` are identical.
## `hp_mdcv` and `hp_cmr` are identical.
## `copd` and `cpd` are identical.
## `dem` and `dementia` are identical.
## `dm` and `diab` are identical.
## `dmc` and `diabwc` are identical.
## `mal` and `canc` are identical.
## `mi_mdcv` and `mi_cmr` are identical.
## `mld_mdcv` and `mld_cmr` are identical.
## `msld_mdcv` and `msld_cmr` are identical.
## `mst` and `metacanc` are identical.
## `pud_mdcv` and `pud_cmr` are identical.
## `pvd_mdcv` and `pvd_cmr` are identical.
## `rhd` and `rheumd` are identical.
## `rnd` and `rend` are identical.
## `cci` and `score` are identical.

```

The only columns remaining in the `mdcv_v_cmr` object are id columns and three columns unique to the return from `medicalcoder::comorbidities()`: * `num_cmr`: the number of flagged comorbidities, * `cmr_flag`: a comorbidity flag (at least one comorbidity), and * `age_score`: an age score (all NA) in this case.

```

str(mdcv_v_cmr)
## Classes 'medicalcoder_comorbidities', 'data.table' and 'data.frame': 545497 obs. of 5 va
## $ subject_id: int 10000032 10000032 10000032 10000032 10000068 10000084 10000084 100001
## $ hadm_id : int 22595853 22841357 25742920 29079034 25022803 23052089 29888819 272509
## $ num_cmr : int 2 2 2 2 0 1 1 0 0 0 ...
## $ cmr_flag : int 1 1 1 1 0 1 1 0 0 0 ...
## $ age_score : int NA NA NA NA NA NA NA NA NA NA ...

```

```
## - attr(*, "sorted")= chr [1:2] "subject_id" "hadm_id"
## - attr(*, ".internal.selfref")=<pointer: 0x105cbfbb0>
```

6.1.2 medicalcoder::comorbidities() vs MIMIC-IV Code

There are some notable structural differences between the results from `medicalcoder::comorbidities()` and the MIMIC-IV Code. First, note that the number of rows differ. This time for different reasons than when we saw different results between `medicalcoder::comorbidities()` and `comorbidity::comorbidity()`.

```
nrow(medicalcoder_charlson_mimiciv)
## [1] 687203
nrow(mimiciv_charlson)
## [1] 546028
```

These records with no comorbidities are not returned by the MIMIC-IV Code. More precisely, all subject/hadm_id pairs missing from the MIMIC-IV Code have a missing `hadm_id`.

```
mimicivDT[!mimiciv_charlson, on = c("hadm_id", "hadm_id")][, all(is.na(hadm_id))]
## [1] TRUE
```

Build the `mdcr_v_mmcc` object to make comparing the results between the `medicalcoder::comorbidities()` call and the MIMIC-IV SQL easier.

```
mdcr_v_mmcc <-
  merge(
    x = medicalcoder_charlson_mimiciv,
    y = mimiciv_charlson,
    all = FALSE,
    by = c("subject_id", "hadm_id"),
    suffixes = c("_mdcr", "_mimiciv")
  )
```

Another difference in the output from the MIMIC-IV SQL and `medicalcoder::comorbidities()` is that when an encounter has ICD codes for the less severe and more severe version of a condition the MIMIC-IV code reports the flag for both whereas `medicalcoder::comorbidities()` only returns the flag for the more severe case. We set the less severe case for the MIMIC-IV code results to zero when the more severe case is present before comparing all the results.

```

# Set the MIMIC-IV code DM without cc to 0 when cc exist
mdcr_v_mmcc[
  diabetes_without_cc == 1L & diabetes_with_cc == 1L,
  diabetes_without_cc := 0L
]

# Set mild liver disease to 0 when sever liver disease exists for MIMIC-IV code
# results
mdcr_v_mmcc[
  mild_liver_disease == 1L & severe_liver_disease == 1L,
  mild_liver_disease := 0L
]

# set malignant_cancer to zero when metastatic_solid_tumor exists for MIMIC-IV
# code results
mdcr_v_mmcc[
  malignant_cancer == 1L & metastatic_solid_tumor == 1L,
  malignant_cancer := 0L
]

```

Now that the flags have been adjusted, the relevant columns to compare between `medicalcoder::comorbidities()` and the MIMIC-IV Code are:

```

mvm_columns <-
  data.table::fread(text = "
    medicalcoder | mimiciivcode
    aidshiv      | aids
    cebvd        | cerebrovascular_disease
    chf           | congestive_heart_failure
    copd          | chronic_pulmonary_disease
    dem           | dementia
    dm            | diabetes_without_cc
    dmc           | diabetes_with_cc
    hp            | paraplegia
    mal           | malignant_cancer
    mi            | myocardial_infarct
    mld           | mild_liver_disease
    msld          | severe_liver_disease
    mst           | metastatic_solid_tumor
    pud           | peptic_ulcer_disease
    pvd           | peripheral_vascular_disease
    rhd           | rheumatic_disease
  ")

```

```

    rnd          | renal_disease
    age_score_mdcr | age_score_mimiciv
    cci          | charlson_comorbidity_index
  ")

```

If the columns are identical, they are removed from the `mdcr_v_mmcc` object.

```

for (i in seq_len(nrow(mvm_columns))) {
  x <- mvm_columns[["medicalcoder"]][i]
  y <- mvm_columns[["mimicivcode"]][i]
  z <- identical(mdcr_v_mmcc[[x]], mdcr_v_mmcc[[y]])
  if (z) {
    message(sprintf("`%s` and `%s` are identical.", x, y))
    data.table::set(mdcr_v_mmcc, j = x, value = NULL)
    data.table::set(mdcr_v_mmcc, j = y, value = NULL)
  } else {
    stop(sprintf("`%s` and `%s` are not identical.", x, y))
  }
}

## `aidshiv` and `aids` are identical.
## `cebud` and `cerebrovascular_disease` are identical.
## `chf` and `congestive_heart_failure` are identical.
## `copd` and `chronic_pulmonary_disease` are identical.
## `dem` and `dementia` are identical.
## `dm` and `diabetes_without_cc` are identical.
## `dmc` and `diabetes_with_cc` are identical.
## `hp` and `paraplegia` are identical.
## `mal` and `malignant_cancer` are identical.
## `mi` and `myocardial_infarct` are identical.
## `mld` and `mild_liver_disease` are identical.
## `msld` and `severe_liver_disease` are identical.
## `mst` and `metastatic_solid_tumor` are identical.
## `pud` and `peptic_ulcer_disease` are identical.
## `pvd` and `peripheral_vascular_disease` are identical.
## `rhd` and `rheumatic_disease` are identical.
## `rnd` and `renal_disease` are identical.
## `age_score_mdcr` and `age_score_mimiciv` are identical.
## `cci` and `charlson_comorbidity_index` are identical.

```

The only remaining columns are id columns and the `num_cmrb` column, which is unique to the return from `medicalcoder::comorbidities()`.

```
str(mdcv_v_mmcc)
## Classes 'medicalcoder_comorbidities', 'data.table' and 'data.frame': 546028 obs. of 4 va
## $ subject_id: int 10000032 10000032 10000032 10000032 10000068 10000084 10000084 100001
## $ hadm_id : int 22595853 22841357 25742920 29079034 25022803 23052089 29888819 272509
## $ num_cmrb : int 2 2 2 2 0 1 1 0 0 0 ...
## $ cmrb_flag : int 1 1 1 1 0 1 1 0 0 0 ...
## - attr(*, "sorted")= chr [1:2] "subject_id" "hadm_id"
## - attr(*, ".internal.selfref")=<pointer: 0x105cbfbb0>
## - attr(*, "index")= int(0)
```

6.2 Elixhauser Comorbidities

6.2.1 medicalcoder::comorbidities() vs comorbidity::comorbidity()

As with the Charlson comparison, the numbers of rows between the two results differ. And for the same reason: no diagnostic ICD codes.

```
nrow(medicalcoder_elixhauser_quan2005)
## [1] 687203
nrow(comorbidity_elixhauser)
## [1] 545497
# subject/hadm_id not in comorbidity::comorbidity() results all have no
# diagnostic codes.
mimicivDT[!comorbidity_elixhauser][, any(dx == 1 & !is.na(icd_code))]
## [1] TRUE
medicalcoder_elixhauser_quan2005[!comorbidity_elixhauser][, all(cmrb_flag == 0L)]
## [1] TRUE
```

We build the `mdcr_v_cmrb_elix` object to make comparing the results between the `medicalcoder::comorbidities()` call and the `comorbidity::comorbidity()` call easier. We will only include the `subject_id/hadm_id` reported on by both.

```
mdcr_v_cmrb_elix <-
  merge(
    x = medicalcoder_elixhauser_quan2005,
    y = comorbidity_elixhauser,
    all = FALSE,
    by = c("subject_id", "hadm_id"),
    suffixes = c("_mdcr", "_cmrb")
  )
```

The relevant columns to compare are as follows:

```
mvc_elix_columns <-  
  data.table::fread(text = "  
    medicalcoder      | comorbidity  
    AIDS              | aids  
    ALCOHOL           | alcohol  
    ANEMDEF           | dane  
    ARTH              | rheumd  
    BLDLOSS           | blane  
    CARDIAC_ARRHYTHMIAS | carit  
    CHF              | chf  
    CHRNLUNG          | cpd  
    COAG              | coag  
    DEPRESS           | depre  
    DM                | diabunc  
    DMCX              | diabc  
    DRUG              | drug  
    HTN_CX            | hypc  
    HTN_UNCX          | hypunc  
    HYPOTHY           | hypothy  
    LIVER             | ld  
    LYMPH             | lymph  
    LYLES            | fed  
    METS              | metacanc  
    NEURO             | ond  
    OBESE             | obes  
    PARA             | para  
    PERIVASC          | pvd  
    PSYCH             | psycho  
    PULMCIRC          | pcd  
    RENLFAIL          | rf  
    TUMOR             | solidtum  
    ULCER             | pud  
    VALVE             | valv  
    WGHTLOSS          | wloss  
  ")
```

If the columns are identical, we remove them from the `mdcr_v_cmrb_elix` object.

```
for (i in seq_len(nrow(mvc_elix_columns))) {  
  x <- mvc_elix_columns[["medicalcoder"]][i]
```

```

y <- mvc_elix_columns[["comorbidity"]][i]
z <- identical(mdcv_v_cmrb_elix[[x]], mdcv_v_cmrb_elix[[y]])
if (z) {
  message(sprintf("`%s` and `%s` are identical.", x, y))
  data.table::set(mdcv_v_cmrb_elix, j = x, value = NULL)
  data.table::set(mdcv_v_cmrb_elix, j = y, value = NULL)
} else {
  stop(sprintf("`%s` and `%s` are not identical.", x, y))
}
}

## `AIDS` and `aids` are identical.
## `ALCOHOL` and `alcohol` are identical.
## `ANEMDEF` and `dane` are identical.
## `ARTH` and `rheumd` are identical.
## `BLDLOSS` and `blane` are identical.
## `CARDIAC_ARRHYTHMIAS` and `carit` are identical.
## `CHF` and `chf` are identical.
## `CHRNLUNG` and `cpd` are identical.
## `COAG` and `coag` are identical.
## `DEPRESS` and `depre` are identical.
## `DM` and `diabunc` are identical.
## `DMCX` and `diabc` are identical.
## `DRUG` and `drug` are identical.
## `HTN_CX` and `hypc` are identical.
## `HTN_UNCX` and `hypunc` are identical.
## `HYPOTHY` and `hypoathy` are identical.
## `LIVER` and `ld` are identical.
## `LYMPH` and `lymph` are identical.
## `LYTES` and `fed` are identical.
## `METS` and `metacanc` are identical.
## `NEURO` and `ond` are identical.
## `OBESE` and `obes` are identical.
## `PARA` and `para` are identical.
## `PERIVASC` and `pvd` are identical.
## `PSYCH` and `psycho` are identical.
## `PULMCIRC` and `pcd` are identical.
## `RENLFAIL` and `rf` are identical.
## `TUMOR` and `solidtum` are identical.
## `ULCER` and `pud` are identical.
## `VALVE` and `valv` are identical.
## `WGHTLOSS` and `wloss` are identical.

```

The only columns remaining in the `mdcr_v_cmrb_elix` object are id columns and columns unique to the return from `medicalcoder::comorbidities()`: * `num_cmrb`: the number of flagged comorbidities, * `cmrb_flag`: a comorbidity flag (at least one comorbidity), * `HTN_C`: any hypertension, * `mortality_index` and `readmission_index`: based on the weights from AHRQ.

```
str(mdcr_v_cmrb_elix)
## Classes 'medicalcoder_comorbidities', 'data.table' and 'data.frame': 545497 obs. of 7 va
## $ subject_id      : int  10000032 10000032 10000032 10000032 10000068 10000084 10000084
## $ hadm_id         : int  22595853 22841357 25742920 29079034 25022803 23052089 29888819
## $ HTN_C           : int    0 0 0 0 0 0 0 0 0 0 ...
## $ num_cmrb        : int    3 4 3 4 1 1 1 0 1 1 ...
## $ cmrb_flag       : int    1 1 1 1 1 1 1 0 1 1 ...
## $ mortality_index : int    2 29 18 27 -1 5 5 0 0 0 ...
## $ readmission_index: int   22 33 26 36 6 7 7 0 0 0 ...
## - attr(*, "sorted")= chr [1:2] "subject_id" "hadm_id"
## - attr(*, ".internal.selfref")=<pointer: 0x105cbfbb0>
```

6.3 PCCC

6.3.1 `medicalcoder::comorbidities()` vs `pccc::ccc()`

As we have seen in other cases, the absence of ICD codes for a `hadm_id` results in the omission of the record from the return from `pccc::ccc()`.

```
nrow(medicalcoder_pccc_v2.0)
## [1] 687203
nrow(pccc_pcccv2.0)
## [1] 545576
```

For the common `subject_id/hadm_id` pairs we compare the flags.

```
mdcr_v_pccc <-
  merge(
    x = medicalcoder_pccc_v2.0,
    y = pccc_pcccv2.0,
    all = FALSE,
    by = c("subject_id", "hadm_id"),
    suffixes = c("_mdcr", "_pccc")
  )
```

```

mvp_columns <- data.table::fread(text = "
medicalcoder | pccc
congeni_genetic_mdcr | congeni_genetic_pccc
cvd_mdcr | cvd_pccc
gi_mdcr | gi_pccc
hemato_immu_mdcr | hemato_immu_pccc
malignancy_mdcr | malignancy_pccc
metabolic_mdcr | metabolic_pccc
neonatal_mdcr | neonatal_pccc
neuromusc_mdcr | neuromusc_pccc
renal_mdcr | renal_pccc
respiratory_mdcr | respiratory_pccc
cmrb_flag | ccc_flag"
)

```

```

for (i in seq_len(nrow(mvp_columns))) {
  x <- mvp_columns[["medicalcoder"]][i]
  y <- mvp_columns[["pccc"]][i]
  z <- identical(mdcr_v_pccc[[x]], mdcr_v_pccc[[y]])
  if (z) {
    message(sprintf("`%s` and `%s` are identical.", x, y))
    data.table::set(mdcr_v_pccc, j = x, value = NULL)
    data.table::set(mdcr_v_pccc, j = y, value = NULL)
  } else {
    stop(sprintf("`%s` and `%s` are not identical.", x, y))
  }
}

## `congeni_genetic_mdcr` and `congeni_genetic_pccc` are identical.
## `cvd_mdcr` and `cvd_pccc` are identical.
## `gi_mdcr` and `gi_pccc` are identical.
## `hemato_immu_mdcr` and `hemato_immu_pccc` are identical.
## `malignancy_mdcr` and `malignancy_pccc` are identical.
## `metabolic_mdcr` and `metabolic_pccc` are identical.
## `neonatal_mdcr` and `neonatal_pccc` are identical.
## `neuromusc_mdcr` and `neuromusc_pccc` are identical.
## `renal_mdcr` and `renal_pccc` are identical.
## `respiratory_mdcr` and `respiratory_pccc` are identical.
## `cmrb_flag` and `ccc_flag` are identical.

```

There are some differences in the results between the two approaches.

```
str(mdc_r_v_pccc)
## Classes 'medicalcoder_comorbidities', 'data.table' and 'data.frame': 545576 obs. of 8 va
## $ subject_id : int 10000032 10000032 10000032 10000032 10000068 10000084 10000084 10
## $ hadm_id : int 22595853 22841357 25742920 29079034 25022803 23052089 29888819 27
## $ misc : int 0 0 1 1 0 0 0 0 0 0 ...
## $ any_tech_dep : int 0 0 1 1 0 0 0 0 0 0 ...
## $ any_transplant: int 0 0 0 0 0 0 0 0 0 0 ...
## $ num_cmr : int 1 2 3 3 0 2 2 0 1 1 ...
## $ tech_dep : int 0 0 1 1 0 0 0 0 0 0 ...
## $ transplant : int 0 0 0 0 0 0 0 0 0 0 ...
## - attr(*, "sorted")= chr [1:2] "subject_id" "hadm_id"
## - attr(*, ".internal.selfref")= <pointer: 0x105cbfbb0>
```

The `misc` column is the “miscellaneous” category reported by `medicalcoder::comorbidities()` and is not reported by `pccc::ccc()`. The existence of the `misc` column and some differences in the returned results between `pccc::ccc()` version 1.0.7, and `medicalcoder::comorbidities()` is due to how `medicalcoder` is implemented.

The differences in any technology dependence and transplant are due to design difference between *medicalcoder* and *pccc*. *medicalcoder* uses database style joins to map precomputed ICD-codes to conditions. This allows for a simple extension of mapping ICD-codes to both the primary condition and subcondition as the latter require only an additional column in a lookup table. The *pccc* uses partial string matching to map ICD codes to conditions for each of the primary conditions and only the two subconditions of technology dependence and transplant. When building *medicalcoder*, the lookup table with condition and subconditions was extended to match the publication documentation.

```
mdc_r_v_pccc[, .N, keyby = .(medicalcoder = any_tech_dep, pccc = tech_dep)]
## Key: <medicalcoder, pccc>
##      medicalcoder pccc      N
##      <int> <int> <int>
## 1:          0      0 457223
## 2:          1      0   615
## 3:          1      1 87738
mdc_r_v_pccc[, .N, keyby = .(medicalcoder = any_transplant, pccc = transplant)]
## Key: <medicalcoder, pccc>
##      medicalcoder pccc      N
##      <int> <int> <int>
## 1:          0      0 525925
## 2:          1      0   4863
## 3:          1      1 14788
```

Several GitHub issues have been created for *pccc* to correct these issues for specific ICD codes:

- ICD-9 349.1
- ICD-9 V56
- ICD-10 Z49
- ICD-9 86.06
- ICD-9 V45.85
- ICD-9 V53.3
- ICD-9 V53.91
- ICD-9 V65.46
- ICD-9 37.52
- ICD-9 V42.0
- ICD-10 Z94

6.4 Summary of *medicalcoder* vs Other Implementations

Using the MIMIC-IV data set we ...

Table X

<i>medicalcoder</i>	Comparator	Agreement summary	Main reason for discordance
<code>charlson_mimicivcode</code>	MIMIC-IV Code	Exact match after harmonizing severity handling	Structural output/severity handling
<code>charlson_quan2011</code>	<i>comorbidity</i>	Exact match for shared flags	Regex/prefix matching vs valid-code lookup
<code>elixhauser_quan2005</code>	<i>comorbidity</i>	Exact match for shared flags	Extra <i>medicalcoder</i> summary/index fields
<code>pccc_v2.0</code>	<i>pccc</i>	High agreement, with documented differences	Technology dependence/transplant mapping

7 Precomputed vs Regular Expressions

The default method in *medicalcoder* for mapping ICD codes to comorbidities is to use a precomputed sets of known ICD codes and use database style joins to and lookup tables. *comorbidity*, and many other utilities, use regular expressions or partial string matching to make the maps. This can lead to problems. For example, the strings “C189N” and “C189NOTACODE” are not a valid ICD code, and will be flagged as a cancer comorbidity via `comorbidity::comorbidity()` but will not be flagged by `medicalcoder::comorbidities()`.

```

DF <-
  data.frame(
    icd_codes = c("C1", "C18", "C189", "C189N", "C189NOTACODE"),
    codeids   = c("C1", "C18", "C189", "C189N", "C189NOTACODE"),
    stringsAsFactors = FALSE
  )
canc <-
  comorbidity::comorbidity(
    x = DF,
    id = "codeids",
    code = "icd_codes",
    map = "charlson_icd10_quan",
    assign0 = TRUE
  )[, c("codeids", "canc")]
mal <-
  medicalcoder::comorbidities(
    data = DF,
    id.var = "codeids",
    icd.codes = "icd_codes",
    method = "charlson_quan2005",
    poa = 1,
    primarydx = 0
  )[, c("codeids", "mal")]

merge(x = canc, mal, by = c("codeids"))
##      codeids canc mal
## 1         C1     0  0
## 2        C18     1  1
## 3       C189     1  1
## 4      C189N     1  0
## 5 C189NOTACODE     1  0

```

There are cases where a regular expressions could be useful. Consider the ICD-10-GM (German version of ICD-10). As of *medicalcoder* version 0.8.1.9002, the ICD-10-GM codes are not part of the internal ICD database.

There are codes in ICD-10-GM which are not in the CDC, CMS, WHO, IHACPA, or Socialstyrelsen systems which are part of the *medicalcoder* ICD database. For example, ICD-10-GM [E11.74](#): Type 2 diabetes mellitus With multiple complications With diabetic foot syndrome, controlled, and [E11.75](#): Type 2 diabetes mellitus With multiple complications With diabetic foot syndrome, uncontrolled. These two codes are extensions under the E11 category; subcategory E11.7 which is listed in Quan et al. (2005) as diabetes with complications for Charlson and Elixhauser.

```

subset(
  x = medicalcoder::get_icd_codes(with.description = TRUE),
  subset = startsWith(code, "E117"),
  select = c("icdv", "dx", "full_code", "code", "src", "desc")
)
##          icdv dx full_code  code          src
## 154640    10  1      E11.7  E117 socialstyrelsen
## 154641    10  1      E11.7  E117          cdc
## 154642    10  1      E11.7  E117          who
## 154643    10  1      E11.7  E117        ihacpa
## 154644    10  1      E11.7  E117          who
## 154645    10  1    E11.70 E1170        ihacpa
## 154646    10  1    E11.71 E1171        ihacpa
## 154647    10  1    E11.72 E1172        ihacpa
## 154648    10  1    E11.73 E1173        ihacpa
##
## 154640                                     Diabetes mellitus typ 2 - med multipla
## 154641                               Non-insulin-dependent diabetes mellitus with multipl
## 154642                               Non-insulin-dependent diabetes mellitus with multipl
## 154643                                   Type 2 diabetes mellitus with multipl
## 154644                                   Type 2 diabetes mellitus with multipl
## 154645   Non-insulin-dependent diabetes mellitus with multiple complications, not stated
## 154646 Type 2 diabetes mellitus with multiple microvascular or other specified nonvascula
## 154647                                   Type 2 diabetes mellitus with features of ins
## 154648                                   Type 2 diabetes mellitus with foot ulcer due to

subset(
  x = medicalcoder::get_charlson_codes(),
  subset = startsWith(code, "E117") & charlson_quan2005 == 1L,
  select = c("icdv", "dx", "full_code", "code", "condition")
)
##          icdv dx full_code  code condition
## 6019    10  1      E11.7  E117         dmc
## 6020    10  1    E11.70 E1170         dmc
## 6021    10  1    E11.71 E1171         dmc
## 6022    10  1    E11.72 E1172         dmc
## 6023    10  1    E11.73 E1173         dmc

subset(
  x = medicalcoder::get_elixhauser_codes(),
  subset = startsWith(code, "E117") & elixhauser_quan2005 == 1L,
  select = c("icdv", "dx", "full_code", "code", "condition")
)

```

```
)
##      icdv dx full_code code condition
## 5554   10  1    E11.7  E117      DMCX
## 5555   10  1    E11.70 E1170      DMCX
## 5556   10  1    E11.71 E1171      DMCX
## 5557   10  1    E11.72 E1172      DMCX
## 5558   10  1    E11.73 E1173      DMCX
```

Thus, if you have a data-set with ICD-10-GM codes, you should, expect false negative flags for diabetes with complications when using the default settings in `medicalcoder::comorbidities()`

```
# data from https://gesund.bund.de/en/icd-code-suche/e11-7
GMdata <- data.table::fread(text = "
CodeID;ICD-Code;Description
ICD-10-GM E11.72;E11.72;Type 2 diabetes mellitus With multiple complications With other mult.
ICD-10-GM E11.73;E11.73;Type 2 diabetes mellitus With multiple complications With other mult.
ICD-10-GM E11.74;E11.74;Type 2 diabetes mellitus With multiple complications With diabetic f
ICD-10-GM E11.75;E11.75;Type 2 diabetes mellitus With multiple complications With diabetic f

common_args <- list( data = GMdata, id.vars = "CodeID", icd.codes = "ICD-Code", dx = 1, icdv

do.call(
  what = medicalcoder::comorbidities,
  args = c(common_args, list(method = "charlson_quan2005"))
)[, c("CodeID", "dm", "dmc")]
##
## Comorbidities via charlson_quan2005
##
##      CodeID      dm      dmc
##      <char> <int> <int>
## 1: ICD-10-GM E11.72      0      1
## 2: ICD-10-GM E11.73      0      1
## 3: ICD-10-GM E11.74      0      0
## 4: ICD-10-GM E11.75      0      0

do.call(
  what = medicalcoder::comorbidities,
  args = c(common_args, list(method = "elixhauser_quan2005"))
)[, c("CodeID", "DM", "DMCX")]
##
## Comorbidities via elixhauser_quan2005
```

```
##
##           CodeID      DM  DMCX
##           <char> <int> <int>
## 1: ICD-10-GM E11.72      0    1
## 2: ICD-10-GM E11.73      0    1
## 3: ICD-10-GM E11.74      0    0
## 4: ICD-10-GM E11.75      0    0
```

If, however, you use `mapping = "regex"` in the call to `medicalcoder::comorbidities()` (new in v0.9.0), then these conditions will be flagged, at least for Charlson, the Elixhauser regex has not yet been implemented.

```
do.call(
  what = medicalcoder::comorbidities,
  args = c(common_args, list(method = "charlson_quan2005", mapping = "regex"))
)[, c("CodeID", "dm", "dmc")]
##
## Comorbidities via charlson_quan2005
##
##           CodeID      dm  dmc
##           <char> <int> <int>
## 1: ICD-10-GM E11.72      0    1
## 2: ICD-10-GM E11.73      0    1
## 3: ICD-10-GM E11.74      0    1
## 4: ICD-10-GM E11.75      0    1

do.call(
  what = medicalcoder::comorbidities,
  args = c(common_args, list(method = "elixhauser_quan2005", mapping = "regex"))
)[, c("CodeID", "DM", "DMCX")]
## Error:
## ! mapping = "regex" for Elixhauser methods has not yet been implemented
```

The current regular expressions are available to end users via

```
str(medicalcoder:::mdcr_internal_charlson_regex..)
## 'data.frame':   117 obs. of  11 variables:
## $ condition      : chr  "aids" "aids" "aidshiv" "aidshiv" ...
## $ icdv           : int   9 10 9 10 10 10 9 9 10 10 ...
## $ dx             : int   1 1 1 1 1 1 1 1 1 1 ...
## $ pattern        : chr   "~(0(0(3\\.?.1/7\\.?[24])|1[0-8]|31|46\\.?.3/54/78\\.?.5)
```

```
## $ charlson_cdmf2019      : int  1 1 0 0 0 0 1 0 0 0 ...
## $ charlson_deyo1992     : int  0 0 1 0 0 0 0 1 0 0 ...
## $ charlson_ludvigsson2021 : int  0 0 0 1 0 0 0 0 0 1 ...
## $ charlson_mimiciuvcde  : int  0 0 1 0 1 0 1 0 0 0 ...
## $ charlson_quan2005     : int  0 0 1 0 1 0 1 0 0 0 ...
## $ charlson_sundararajan2004 : int  0 0 0 0 0 1 0 0 1 0 ...
## $ charlson_quan2011     : int  0 0 1 0 1 0 1 0 0 0 ...
```

The regular expression patterns have been written to match both full and compact ICD codes.

Revisiting the first example in this section, we can see that using `mapping = "regex"` in `medicalcoder::comorbidities()` will give the same false-positives as `comorbidity::comorbidity()` does. The utility exists, but it should be used with care.

```
mal2 <-
  medicalcoder::comorbidities(
    data = DF,
    id.var = "codeids",
    icd.codes = "icd_codes",
    method = "charlson_quan2005",
    poa = 1,
    primarydx = 0,
    mapping = "regex"
  )[, c("codeids", "mal")]

merge(
  x = merge(x = canc, mal, by = c("codeids")),
  y = mal2,
  by = c("codeids"),
  suffixes = c("_precomputed", "_by_regex")
)

##           codeids  canc  mal_precomputed  mal_by_regex
## 1             C1      0                0              0
## 2             C18      1                1              1
## 3            C189      1                1              1
## 4           C189N      1                0              1
## 5 C189NOTACODE      1                0              1
```

8 Longitudinal Flagging

Under the assumption that all relevant ICD codes are (re)reported on each encounter then there is no need to carry-forward comorbidity flags. However, that assumption is rarely valid. *medicalcoder* provides a utility for indefinite carry-forward of a condition. As reported in the primary manuscript this includes accounting for conditions which should not be reported on a specific encounter but are reported on the following encounters. In the following we will explore some of the pros and cons of the indefinite carry-forward of comorbidities.

To simplify the work we will reapply the comorbidity algorithms using `enc_seq` instead of `hadm_id` so that the encounters are considered in chronologic order. We present here the code for Charlson, Elixhauser, and PCCC, which will be referenced several times within the remainder of this document.

```
medicalcoder_charlson_current <-  
  medicalcoder::comorbidities(  
    data      = mimiciuDT,  
    icd.codes = "icd_code",  
    id.vars   = c("subject_id", "enc_seq"),  
    icdv.var  = "icd_version",  
    dx.var    = "dx",  
    poa       = 1L,  
    primarydx = 0L,  
    method    = "charlson_quan2011",  
    flag.method = "current"  
  )
```

```
medicalcoder_charlson_cumulative <-  
  medicalcoder::comorbidities(  
    data      = mimiciuDT,  
    icd.codes = "icd_code",  
    id.vars   = c("subject_id", "enc_seq"),  
    icdv.var  = "icd_version",  
    dx.var    = "dx",  
    poa       = 1L,  
    primarydx = 0L,  
    method    = "charlson_quan2011",  
    flag.method = "cumulative"  
  )
```

```
medicalcoder_elixhauser_current <-  
  medicalcoder::comorbidities(  
    data      = mimiciuDT,  
    icd.codes = "icd_code",  
    id.vars   = c("subject_id", "enc_seq"),  
    icdv.var  = "icd_version",  
    dx.var    = "dx",  
    poa       = 1L,  
    primarydx = 0L,  
    method    = "elixhauser",  
    flag.method = "current"
```

```

data      = mimicivDT,
icd.codes = "icd_code",
id.vars   = c("subject_id", "enc_seq"),
icdv.var  = "icd_version",
dx.var    = "dx",
poa       = 1L,
primarydx = 0L,
method    = "elixhauser_quan2005",
flag.method = "current"
)

```

```

medicalcoder_elixhauser_cumulative <-
  medicalcoder::comorbidities(
    data      = mimicivDT,
    icd.codes = "icd_code",
    id.vars   = c("subject_id", "enc_seq"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    poa       = 1L,
    primarydx = 0L,
    method    = "elixhauser_quan2005",
    flag.method = "cumulative"
  )

```

```

medicalcoder_pcccv2.0_current <-
  medicalcoder::comorbidities(
    data      = mimicivDT,
    icd.codes = "icd_code",
    id.vars   = c("subject_id", "enc_seq"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    poa       = 1L,
    method    = "pccc_v2.0",
    flag.method = "current"
  )

```

```

medicalcoder_pcccv2.0_cumulative <-
  medicalcoder::comorbidities(
    data      = mimicivDT,
    icd.codes = "icd_code",
    id.vars   = c("subject_id", "enc_seq"),
    icdv.var  = "icd_version",

```

```

    dx.var      = "dx",
    poa         = 1L,
    method      = "pccc_v2.0",
    flag.method = "cumulative"
  )

```

```

medicalcoder_pcccv3.1_current <-
  medicalcoder::comorbidities(
    data      = mimicivDT,
    icd.codes = "icd_code",
    id.vars   = c("subject_id", "enc_seq"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    poa       = 1L,
    method    = "pccc_v3.1",
    flag.method = "current"
  )

```

```

medicalcoder_pcccv3.1_cumulative <-
  medicalcoder::comorbidities(
    data      = mimicivDT,
    icd.codes = "icd_code",
    id.vars   = c("subject_id", "enc_seq"),
    icdv.var  = "icd_version",
    dx.var    = "dx",
    poa       = 1L,
    method    = "pccc_v3.1",
    flag.method = "cumulative"
  )

```

8.1 Severity of Disease

Let's look at the impact of longitudinal flagging vs encounter level flagging of Charlson comorbidities.

Figure 1 shows the number of encounters flagged with a given condition for `flag.method = 'current'` and `flag.method = 'cumulative'` for the Charlson condition flagged by `medicalcoder::comorbidities()`. For most conditions the number of encounters flagged is greater when using the cumulative flag method compared to the the current flag method. However, there is a notable counter example: dm (diabetes without complications) and dmc (diabetes with complications). *medicalcoder* will carry forward the severity of disease and report the more severe version after it has been reported.

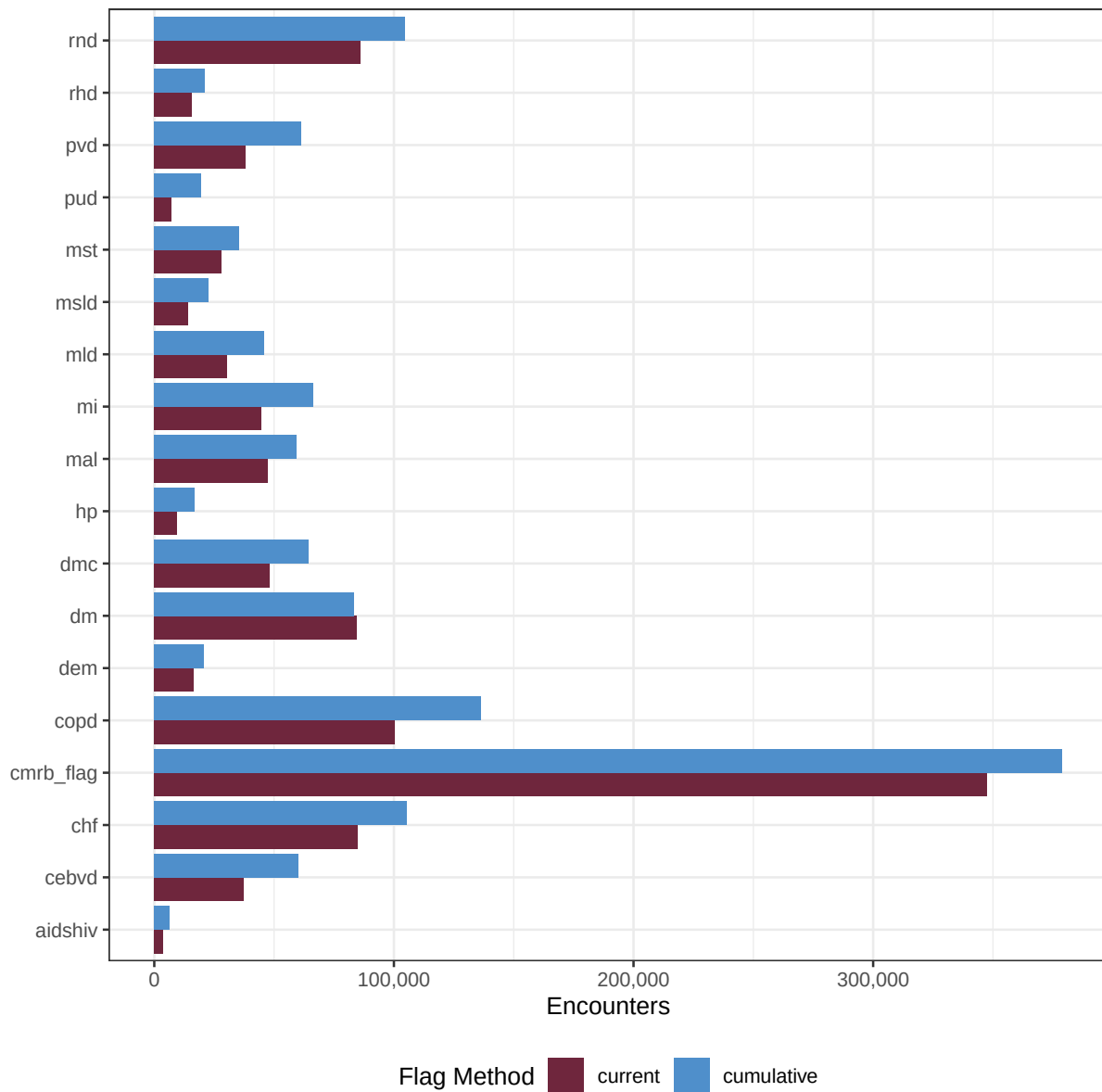


Figure 1: Number of encounters flagged with a Charlson comorbidities based on encounter only (`flag.method = "current"`) versus the whole patient history (`flag.method = "cumulative"`).

Table 3: Flagging Diabetes under Charlson comorbidities for MIMIC-IV subject_id 10009326 using `medicalcoder::comorbidities()` `flag.method = "current"` and `flag.method = "cumulative"`.

Encounter	Current		Cumulative	
	Complicated	Uncomplicated	Complicated	Uncomplicated
1		X		X
2				X
3		X		X
4		X		X
5				X
6		X		X
7		X		X
8		X		X
9				X
10		X		X
11				X
12	X		X	
13		X	X	
14		X	X	
15		X	X	
16		X	X	
17		X	X	

An example of a patient record where on an encounter level flagging approach diabetes will be missed entirely and the severity is inaccurate is in Table 3. Subject 10009326 had 17 encounters. At least one ICD code mapping to uncomplicated diabetes under the Charlson *a la* Quan et al. (2005) appeared in 12 encounters, and 1 encounter for complicated diabetes. Under an encounter-level only flagging of comorbidities, this patient would erroneously not have diabetes of any kind on 4 encounters. Using the longitudinal flagging in `medicalcoder::comorbidities()` via `flag.method = 'cumulative'`, we have diabetes flagged on all encounters, and the severity correctly marked for encounters later in the patient record.

8.1.1 Elixhauser

For Elixhauser, Figure 2, shows similar behavior for diabetes as Charlson.

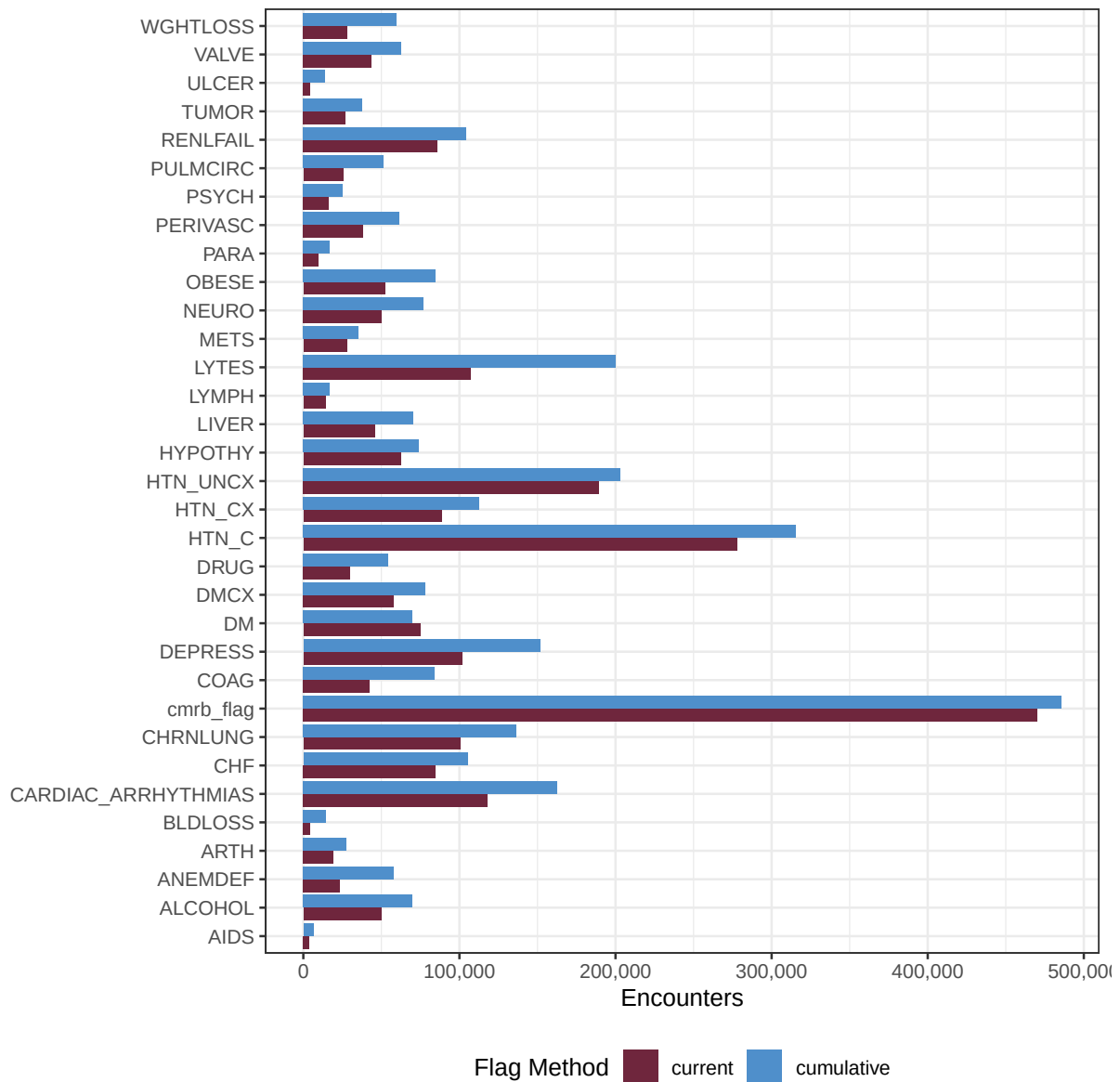


Figure 2: Number of encounters flagged with a Elixhauser comorbidities based on encounter only (`flag.method = current`) verses the whole patient history (`flag.method = "cumulative"`).

8.1.2 PCCC

Figure 3 and Figure 4 illustrate the impact of the flagging method when identifying PCCC comorbidities under PCCC v2 and PCCC v3 respectively. Figure 4 is of particular interest as the implications of technology-dependent codes in PCCC v3 are handled have considerable impact on the flagged comorbidities.

Let's look at the records for `subject_id = 10728333`. Table 4 shows the flagging of metabolic, miscellaneous, and respiratory conditions for PCCC v3 using `flag.method = 'current'` or `flag.method = 'cumulative'`. For this patient, on encounter 1, a technology-dependent ICD code related to a respiratory comorbidity was observed in the patient record. No other ICD codes related to a PCCC v3 comorbidity were observed for encounter 1. As such, no comorbidities are flagged for encounter 1 as at least one non-technology-dependent ICD code is required for any comorbidity to be flagged. Encounter 2 had no PCCC v3 related ICD codes reported. Encounter 3 had a technology-dependent ICD code for a miscellaneous comorbidity reported and again, because no non-technology-dependent ICD was reported no comorbidity is flagged. Starting on encounter 4, a non-technology-dependent ICD code for a metabolic comorbidity was reported. Under the `flag.method = 'current'` that is the only code that is reported and thus only one comorbidity is reported. Under `flag.method = 'cumulative'` the technology-dependent ICD codes from encounters 1 and 3 have been carried forward in the record to encounter 4 and are thus flagged. If the end user were to try to build the cumulative comorbidities from the results of the `flag.method = 'current'` results, the end user would never capture the miscellaneous comorbidity, and would fail to capture the respiratory comorbidity until encounter 6, and would incorrectly classify as non-technology-dependent until the final encounter.

8.2 Diabetes - Chronic vs Acute Disease

Non-gestational diabetes is generally considered a chronic condition. Even when well managed, it is still a relevant diagnosis. The ICD-10 code defining diabetes (with and without complications) for Charlson comorbidities as defined by Quan et al. (2005, 2011), are all in the E10, E11, E12, E13, and E14 categories.

```
subset(  
  x = medicalcoder::get_charlson_codes(),  
  subset = charlson_quan2005 == 1L & condition %in% c("dm", "dmc") & icdv == 10,  
)[["code"]] |>  
substring(1, 3) |>  
unique() |>  
sort()  
## [1] "E10" "E11" "E12" "E13" "E14"
```

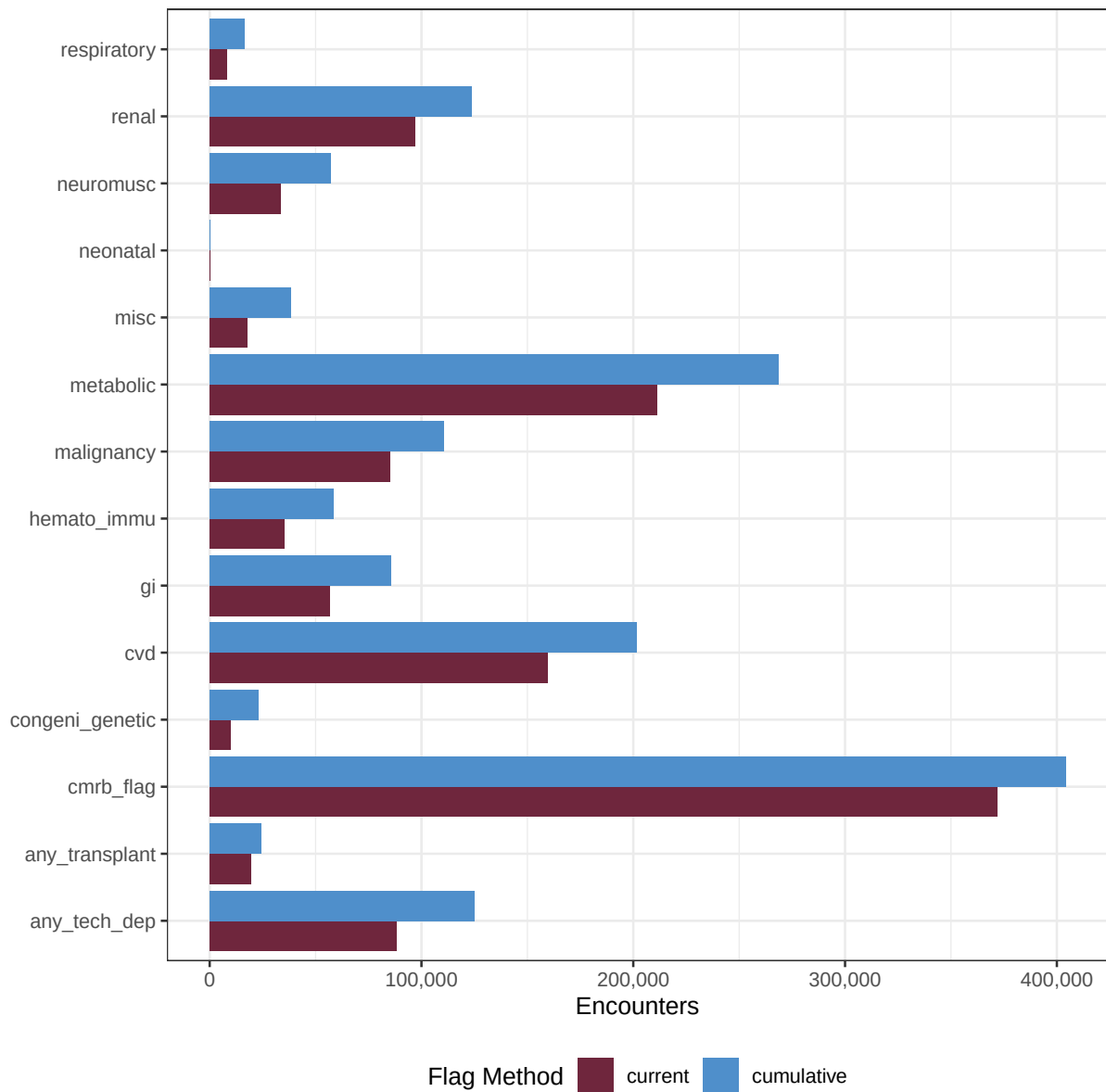


Figure 3: Number of encounters flagged with a PCCC v2 comorbidities based on encounter only (`flag.method = current`) verses the whole patient history (`flag.method = "cumulative"`).

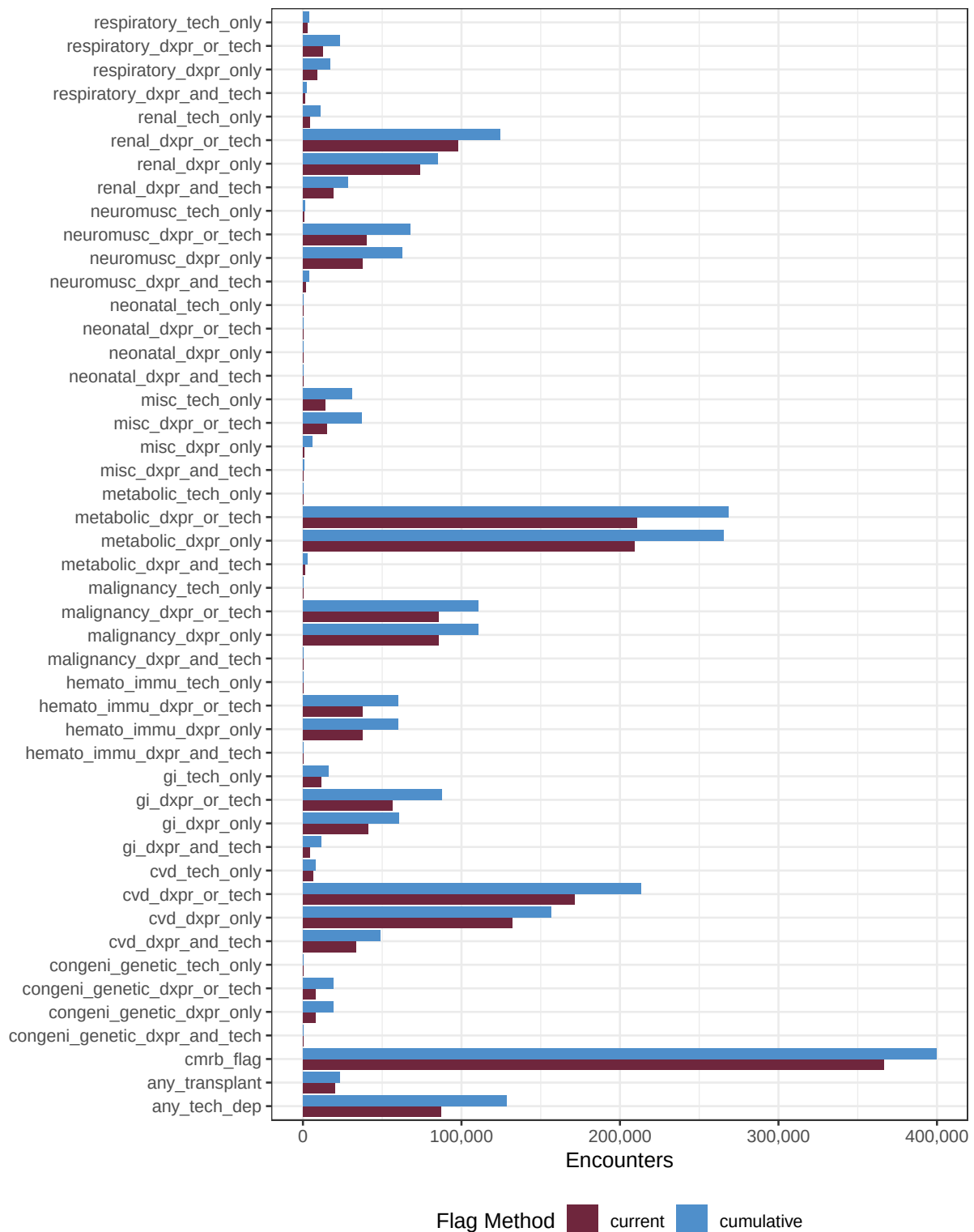


Figure 4: Number of encounters flagged with a PCCC v3 comorbidities based on encounter only (`flag.method = current`) verses the whole patient history (`flag.method = "cumulative"`).

Table 4

Table 5: Flagging of comorbidities under PCCC v3 for MIMIC-IV subject 10728333. Cells marked with DxPr denote that the comorbidity was flagged due to a ICD code that is not technology-dependent. Tech denotes flagging of a comorbidity base on the presence of at least one technology-dependent ICD code and the presence of at least one non-technology-dependent code for another comorbidity. DxPrTech denotes flagging due to both technology-dependent and non-technology-dependent ICD codes. Under the Any Tech columns we mark the encounters where some tech dependence is identified. For this patient, on encounter 1, a technology-dependent ICD code for a respiratory comorbidity was found in the subject’s record. Because no non-technology-dependent code was observed on encounter 1, under PCCC v3 this subject has no comorbidities for encounter 1. This persists for encounters 2 and 3 with respect to respiratory and on encounter 3 for miscellaneous as well. On encounter 4 a non-technology-dependent code for a metabolic comorbidity was reported. Under a cumulative flagging paradigm the miscellaneous and respiratory comorbidities are also flagged as the reported codes from encounters 3 and 1 have been carried forward respectively. Under the ‘current’ flagging method miscellaneous is never flagged as the technology-dependent code never occurs on the same encounter as the non-technology-dependent code.

Encounter	Metabolic		Miscellaneous		Respiratory		Any Tech		Number of Comorbidities	
	Current	Cumulative	Current	Cumulative	Current	Cumulative	Current	Cumulative	Current	Cumulative
1					*	*			0	0
2						†			0	0
3			*	*		†			0	0
4	DxPr	DxPr		Tech		Tech		1	1	3
5	DxPr	DxPr		Tech		Tech		1	1	3
6	DxPr	DxPr		Tech	DxPr	DxPrTech		1	2	3
7	DxPr	DxPr		Tech		DxPrTech		1	1	3
8	DxPr	DxPr		Tech		DxPrTech		1	1	3
9	DxPr	DxPr		Tech		DxPrTech		1	1	3
10	DxPr	DxPr		Tech		DxPrTech		1	1	3
11		DxPr		Tech		DxPrTech		1	0	3
12		DxPr		Tech		DxPrTech		1	0	3
13	DxPr	DxPr		Tech	Tech	DxPrTech	1	1	2	3

* A technology-dependent code was reported on this encounter. Since no non-technology-dependent was reported no comorbidities are flagged.

† A technology-dependent code has been carried forward in the record. Since no non-technology-dependent was reported on this, or a prior, encounter, no comorbidities are flagged.

Let's look at the records for those who have at least two encounters and at least one of the encounters was flagged with diabetes (with or without complications).

```
sids <-
  medicalcoder_charlson_current[
    ,
    .(encounters = length(unique(enc_seq)), any_dm = sum(dm + dmc)),
    by = .(subject_id)
  ][
    encounters > 1 & any_dm > 0,
    subject_id
  ]

dms <-
  merge(
    x = medicalcoder_charlson_current[
      subject_id %in% sids,
      .(subject_id, enc_seq, dm_current = dm, dmc_current = dmc)
    ],
    y = medicalcoder_charlson_cumulative[
      subject_id %in% sids,
      .(subject_id, enc_seq, dm_cumulative = dm, dmc_cumulative = dmc)
    ],
    all = TRUE,
    by = c("subject_id", "enc_seq")
  )
data.table::setDT(dms)
dms[, current := (dm_current + dmc_current)]
dms[, cumulative := (dm_cumulative + dmc_cumulative)]
dms[, total_encs := max(enc_seq), by = .(subject_id)]
dms
## Key: <subject_id, enc_seq>
##      subject_id enc_seq dm_current dmc_current dm_cumulative dmc_cumulative
##           <int>  <int>    <int>      <int>         <int>         <int>
## 1: 10000635      1        1          0            1            0
## 2: 10000635      2        1          0            1            0
## 3: 10000980      1        1          0            1            0
## 4: 10000980      2        1          0            1            0
## 5: 10000980      3        1          0            1            0
## ---
## 146739: 19999287      3        1          0            1            0
## 146740: 19999379      1        1          0            1            0
## 146741: 19999379      2        1          0            1            0
```

```
## 146742: 19999828 1 1 0 1 0
## 146743: 19999828 2 1 0 1 0
##      current cumulative total_encs
##      <int>      <int>      <int>
## 1:      1      1      2
## 2:      1      1      2
## 3:      1      1      7
## 4:      1      1      7
## 5:      1      1      7
## ---
## 146739: 1      1      3
## 146740: 1      1      2
## 146741: 1      1      2
## 146742: 1      1      2
## 146743: 1      1      2
```

How many cumulative flags are on encounters were DM i(with or without complications) is flagged on both a preceding and a preceding encounter, or only after a preceding encounter.

```
dms[, current_flags_before := as.integer((cumsum(current) - current) > 0), by = .(subject_id)]
dms[, current_flags_after  := as.integer(rev(cumsum(rev(current)) - current) > 0), by = .(subject_id)]

dms[, cumulative_between_current_flags := as.integer((current != cumulative) & current_flag)]
dms[, cumulative_only_after_current_flag := as.integer((current != cumulative) & current_flag)]

# Number of Encounters:
dms[, .(Encounters = .N), keyby = .(cumulative_between_current_flags, cumulative_only_after_current_flag)]
## Key: <cumulative_between_current_flags, cumulative_only_after_current_flag>
##      cumulative_between_current_flags cumulative_only_after_current_flag
##                                     <int>                                <int>
## 1:                                     0                                    0
## 2:                                     0                                    1
## 3:                                     1                                    0
##      Encounters
##      <int>
## 1:      131749
## 2:       9219
## 3:       5775
```

At a subject-level, the

```
# counts of subjects with encounters flagged
dms[
  ,
  .(
    between = any(cumulative_between_current_flags),
    after = any(cumulative_only_after_current_flag)
  ),
  by = .(subject_id)
][
  ,
  .(Subjects = .N),
  keyby = .(between, after)
]
## Key: <between, after>
##   between after Subjects
##   <lgcl> <lgcl>   <int>
## 1:  FALSE  FALSE   21241
## 2:  FALSE   TRUE   3267
## 3:   TRUE  FALSE   2086
## 4:   TRUE   TRUE    462
```

Longitudinal flagging of a chronic condition could be beneficial, especially in the cases where we see the condition flagged on encounters preceding and proceeding a specific encounter.

End users are cautioned to understand the specific conditions and comorbidity methods being used before applying the indefinite carry-forward. For example, gestational diabetes is part of the comorbidity mappings for the AHRQ Elixhauser variants based on ICD-10 codes. Note that ICD-10 category E08, E09, E10, E11, E13, and O24 are all diabetes codes with O24 codes mapping to uncomplicated diabetes.

```
ahrq_icd10_dm_codes <-
  subset(
    x = medicalcoder::get_elixhauser_codes(),
    subset = elixhauser_ahrq_icd10 == 1 & condition %in% c("DIAB_CX", "DIAB_UNCX"),
    select = c("code", "condition")
  )
data.table::setDT(ahrq_icd10_dm_codes)
ahrq_icd10_dm_codes[, category := substring(code, 1, 3)]

ahrq_icd10_dm_codes[, .N, keyby = .(category, condition)] |>
  data.table::dcast(category ~ condition, value.var = "N", fill = 0L)
## Key: <category>
```

```
##      category DIAB_CX DIAB_UNCX
##      <char>   <int>   <int>
## 1:      E08      89      5
## 2:      E09      89      5
## 3:      E10      89      3
## 4:      E11      89      6
## 5:      E13      89      5
## 6:      O24       0     42
```

In the MIMIC-IV data, there are subjects with both gestational and non-gestational diabetes, subjects with only gestational diabetes, and subjects with only non-gestational diabetes.

```
DT <- mimicivDT[icd_code %in% ahrq_icd10_dm_codes[condition == "DIAB_UNCX", code]]
DT[, Gestational := any(startsWith(icd_code, "O")), by = .(subject_id)]
DT[, `Non-Gestational` := any(startsWith(icd_code, "E")), by = .(subject_id)]
DT[
  , unique(.SD), .SDcols = c("subject_id", "Gestational", "Non-Gestational")
][
  , .(Subjects = .N), keyby = .(Gestational, `Non-Gestational`)
]
## Key: <Gestational, Non-Gestational>
##      Gestational Non-Gestational Subjects
##      <lgcl>      <lgcl>      <int>
## 1:      FALSE      TRUE      16785
## 2:      TRUE       FALSE       725
## 3:      TRUE       TRUE       139
```

Since the AHRQ variant of Elixhauser comorbidities include both chronic and non-chronic versions of diabetes `flag.method = "cumulative"` may results false positives of diabetes.

8.3 Metastatic Tumors

Metastatic tumors could be consider non-chronic conditions as the tumors could be removed via surgery or treated with other therapies. As such, the flags reported via `flag.method = "cumulative"` for malignancy (Charlson and PCCC), or solid tumors (Elixhauser) should not be interpreted as “present disease” but as “a history of *observed* disease.”

We empathize *observed disease* because there are ICD codes specifically for personal history of malignancies. We collect those ICD codes and the codes mapping to malignancies per Quan et al. (2005, 2011) and subset the `mimicivDT` to subjects with these codes.

```

personal_history_of_malignant <-
  subset(
    x = medicalcoder::get_icd_codes(with.description = TRUE),
    subset = src == "cms" & grepl("personal history of malignant", desc, ignore.case = TRUE)
    select = c("code", "icdv", "dx")
  )
personal_history_of_malignant[["history_of"]] <- 1L

charlson_quan2011_mal <-
  subset(
    x = medicalcoder::get_charlson_codes(),
    subset = condition == "mal" & charlson_quan2011 == 1L,
    select = c("code", "icdv", "dx")
  )
charlson_quan2011_mal[["observed_mal"]] <- 1L

mal <-
  merge(
    x = mimicivDT[!is.na(icd_code), .(subject_id, enc_seq, icd_code, icd_version, dx)],
    y = personal_history_of_malignant,
    all.x = TRUE,
    by.x = c("icd_code", "icd_version", "dx"),
    by.y = c("code", "icdv", "dx")
  )

mal <-
  merge(
    x = mal,
    y = charlson_quan2011_mal,
    all.x = TRUE,
    by.x = c("icd_code", "icd_version", "dx"),
    by.y = c("code", "icdv", "dx")
  )
data.table::setkey(mal, subject_id, enc_seq)

# set flag for any history or observation of a malignancy
mal <-
  mal[
    ,
    .(
      enc_history_of = as.integer(any(history_of, na.rm = TRUE)),
      enc_observed_mal = as.integer(any(observed_mal, na.rm = TRUE))
    )
  ]

```

```

    ),
    by = .(subject_id, enc_seq)
  ]
mal[, subject_history_of := as.integer(sum(enc_history_of) > 0), by = .(subject_id)]
mal[, subject_observed_mal := as.integer(sum(enc_observed_mal) > 0), by = .(subject_id)]

mal <- mal[subject_history_of + subject_observed_mal > 0]

# join on the cumulative flags from medicalcoder for both malignancy and
# metastatic solid tumors. An encounter with both a malignancy and a metastatic
# solid tumor will have the malignancy flag set to zero and retain only the more
# severe condition.
mal <-
  merge(
    x = mal,
    y = medicalcoder_charlson_cumulative[, .(subject_id, enc_seq, medicalcoder_cummulative_malignancy)],
    all.x = TRUE,
    by = c("subject_id", "enc_seq")
  )

mal[, .(Encounters = .N), keyby = .(enc_observed_mal, enc_history_of)]
## Key: <enc_observed_mal, enc_history_of>
##   enc_observed_mal enc_history_of Encounters
##           <int>         <int>         <int>
## 1:                0             0         40443
## 2:                0             1         47083
## 3:                1             0         58910
## 4:                1             1          8222

```

The next chunk will count the number of subjects with reported history of a malignancy and those with an observed malignancy

```

subject_history <-
  mal[
    ,
    lapply(lapply(.SD, any), as.integer),
    .SDcols = c("subject_history_of", "subject_observed_mal"),
    by = .(subject_id)
  ]
subject_history[
  ,
  .(subjects = .N), keyby = .(subject_history_of, subject_observed_mal)
]

```

```

]
## Key: <subject_history_of, subject_observed_mal>
##      subject_history_of subject_observed_mal subjects
##      <int>              <int>              <int>
## 1:              0              1      18127
## 2:              1              0      15229
## 3:              1              1       9360

```

What is notable in the above is that there are 15,229 (35.65%) patients who have only a personal history of malignancy in the data set. If the cumulative flag was to be considered a “history of” these would patients would be considered a false negative. The, the `flag.method = 'cumulative'` can only be interpreted as “current or history of observed disease.”

8.4 AHRQ (2022+) Example - POA

AHRQ Elixhauser for years 2022 and beyond use ICD-10 diagnostic codes and require some ICD-codes to be present-on-admission for the comorbidity to be flag. The MIMIC data does not have a present-on-admission flag. For this example let’s assume that all codes are not POA.

```

common_args <-
  list(
    data = mimiciVDT,
    id.vars = c("subject_id", "enc_seq"),
    icd.codes = "icd_code",
    icdv.var = "icd_version",
    dx.var = "dx",
    poa = 0L,
    primarydx = 0L,
    method = "elixhauser_ahrq_icd10"
  )

flgcurrent <-
  do.call(
    medicalcoder::comorbidities,
    c(common_args, list(flag.method = "current"))
  )

flgcumulative <-
  do.call(
    medicalcoder::comorbidities,

```

```
c(common_args, list(flag.method = "cumulative"))
)
```

Let's look at subject_id 19997538. This subject has ICD code K76.0 reported on encounter 1.

```
mimicivDT[subject_id == 19997538 & icd_code == "K760"]
## Key: <subject_id, enc_seq, hadm_id>
##   subject_id hadm_id   dx seq_num icd_code icd_version chartdate   age
##   <int>      <int> <int>   <int>   <char>      <int>   <IDat>   <num>
## 1:   19997538 22701415    1     15    K760          10    <NA> 53.33048
##   admittime enc_seq      cid
##   <POSct>   <int>      <char>
## 1: 2168-05-01      1 19997538__22701415
```

K76.0 maps to mild liver disease and is required to be POA for the condition to be flagged.

```
subset(medicalcoder::get_icd_codes(with.description = TRUE), full_code == "K76.0")
##   icdv dx full_code code      src known_start known_end
## 177671 10 1    K76.0 K760      cms      2014      2026
## 177672 10 1    K76.0 K760      cdc      2001      2025
## 177673 10 1    K76.0 K760      ihacpa    2020      2026
## 177674 10 1    K76.0 K760      who      2008      2021
## 177675 10 1    K76.0 K760 socialstyrelsen 1997      2026
##   assignable_start assignable_end
## 177671      2014      2026
## 177672      2001      2025
## 177673      2020      2026
## 177674      2008      2021
## 177675      1997      2026
##                                     desc desc_start desc_end
## 177671 Fatty (change of) liver, not elsewhere classified      2014      2026
## 177672 Fatty (change of) liver, not elsewhere classified      2001      2025
## 177673 Fatty (change of) liver, not elsewhere classified      2020      2026
## 177674 Fatty (change of) liver, not elsewhere classified      2008      2021
## 177675 Fettlever som ej klassificeras på annan plats      1997      2026
subset(medicalcoder::get_elixhauser_codes(), full_code == "K76.0")
##   icdv dx full_code code poaexempt condition elixhauser_ahrq_web
## 9300 10 1    K76.0 K760      NA      LIVER      0
## 9301 10 1    K76.0 K760      0 LIVER_MLD      NA
##   elixhauser_elixhauser1988 elixhauser_quan2005 elixhauser_ahrq2022
```

```
## 9300          0          1          NA
## 9301          NA          NA          1
##      elixhauser_ahrq2023 elixhauser_ahrq2024 elixhauser_ahrq2025
## 9300          NA          NA          NA
## 9301          1          1          1
##      elixhauser_ahrq2026 elixhauser_ahrq_icd10
## 9300          NA          NA
## 9301          1          1
subset(medicalcoder::get_elixhauser_poa(), condition == "LIVER_MLD")
##      condition poa_required elixhauser_ahrq2022 elixhauser_ahrq2023
## 21 LIVER_MLD          1          1          1
##      elixhauser_ahrq2024 elixhauser_ahrq2025 elixhauser_ahrq2026
## 21          1          1          1
##      elixhauser_ahrq_icd10
## 21          1
```

When `flag.method = "current"` we see what we expect, mild liver disease is not flagged on any encounter. When `flag.method = "cumulative"` we see that on encounter 1 the condition is not flagged, as expected since it is not POA, but it is flagged on encounters 2 and 3 as the patient history is accounted for.

```
merge(
  x = flgcurrent,
  y = flgcumulative,
  by = c("subject_id", "enc_seq"),
  suffixes = c("_current", "_cumulative")
)[
  subject_id == 19997538,
  .SD,
  .SDcols = patterns("^ (subject_id|enc_seq|LIVER_MLD_c)")
]
## Key: <subject_id, enc_seq>
##      subject_id enc_seq LIVER_MLD_current LIVER_MLD_cumulative
##      <int>      <int>          <int>          <int>
## 1:  19997538      1              0              0
## 2:  19997538      2              0              1
## 3:  19997538      3              0              1
```

Table 6: Time, in seconds, median and IQR, to flag comorbidities via Charlson, Elixhauser, PCCC v2, and PCCC v3 and several different utilities in the MIMIC-IV v3.1 dataset. The flagging method, current or cumulative is reported. Each method was evaluated 10 times.

	<code>medicalcoder::comorbidities()</code>	<code>comorbidity::comorbidity()</code>	MIMIC-IV Code	<code>pccc::ccc()</code>
charlson				
current	0.80 (0.76, 0.85)	16.79 (16.69, 16.88)	34.74 (31.31, 36.55)	
cumulative	1.68 (1.67, 1.70)			
elixhauser				
current	1.24 (1.19, 1.33)	26.74 (26.52, 26.89)		
cumulative	3.52 (3.47, 3.59)			
pcccv2.0				
current	0.92 (0.86, 1.07)			71.96 (71.69, 72.24)
cumulative	2.54 (2.38, 2.69)			
pcccv3.1				
current	1.45 (1.37, 1.57)			
cumulative	3.15 (3.10, 3.25)			

9 Time Required To Compute

The times required (median and IQR) to apply the different comorbidity algorithms to the MIMIC-IV data are shown in Table 6. To process the 7,365,770 total rows of data for 364,627 unique subjects and a total of 687,203 unique hospital encounters, `medicalcoder::comorbidities()` is considerably faster than all the alternatives.

For this work we will include several additional calls to `medicalcoder::comorbidities()`.

The benchmarking is done using *microbenchmark* (Mersmann 2024) and calling each expression ten times.

Figure 5 shows the median time in seconds along with error bars for the IQR for the ten iterations. The reason for the drastic speed improvement between *medicalcoder* and other utilities is that *medicalcoder* uses precomputed maps between known ICD codes and conditions where as *comorbidity* uses regular expressions, and *pccc* uses partical string matching. The MIMIC-IV code also partical string matching and lexigraphical between statements. The numeric values represented in Figure 5 are reported in Table 6.

Saving 7 x 7 in image

10 Summarizing `medicalcoder_comorbidities` Objects

The *medicalcoder* package provides an S3 summary method for the objects returned by `medicalcoder::comorbidities()`. The summaries differ slightly depending of the comorbidity

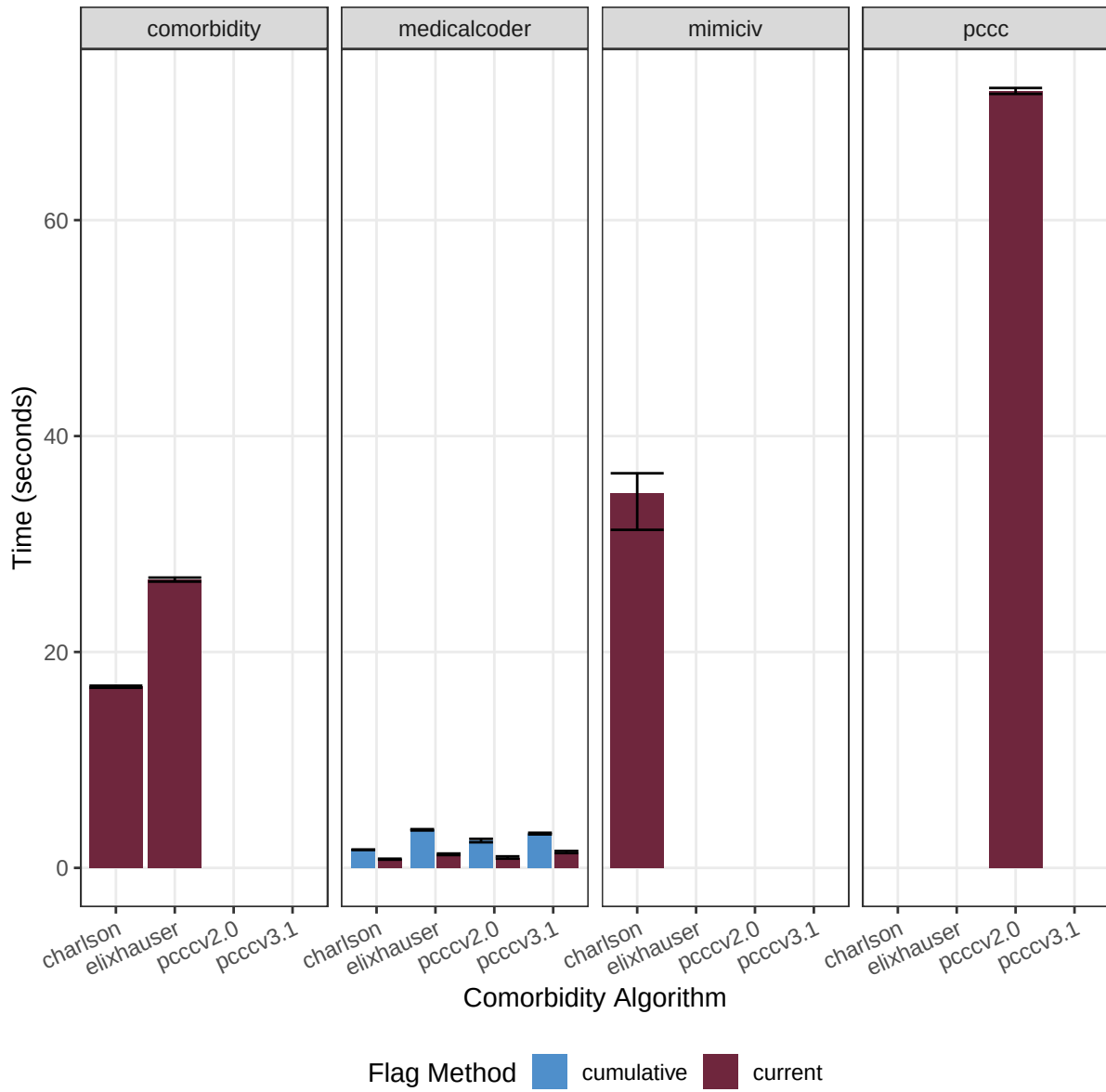


Figure 5: Time (bar height = median, error bars = IQR) in seconds to apply several comorbidity algorithms to the MIMIC-IV dataset. Each method was evaluated 10 times.

algorithm (method). Charlson and Elixhauser return lists with multiple summaries and the PCCC return is a data.frame of summaries.

When the `flag.method = 'current'`, the `conditions` element of the summary for the Charlson and Elixhauser comorbidities is a data.frame with counts and percentages of the distinct `id.vars` with a given comorbidity, and counts of comorbidities. For PCCC v2 and v3, the return is only this data.frame. When the flag method is cumulative, a warning is given stating that the summary may not be meaningful as it is an encounter level summary.

```
str(summary(medicalcoder_charlson_current), max.level = 1)
## List of 3
## $ conditions : 'data.frame': 26 obs. of 4 variables:
## $ age_summary : 'data.frame': 1 obs. of 3 variables:
## $ index_summary: 'data.frame': 1 obs. of 5 variables:
str(summary(medicalcoder_elixhauser_current), max.level = 1)
## List of 2
## $ conditions : 'data.frame': 46 obs. of 3 variables:
## $ index_summary: 'data.frame': 2 obs. of 6 variables:
str(summary(medicalcoder_pcccv2.0_current), max.level = 1)
## 'data.frame': 24 obs. of 4 variables:
## $ condition: chr "congeni_genetic" "cvd" "gi" "hemato_immu" ...
## $ label : chr "Other Congenital or Genetic Defect" "Cardiovascular" "Gastrointestina
## $ count : int 9960 159359 56797 35395 85183 211049 17549 39 33640 97078 ...
## $ percent : num 1.45 23.19 8.26 5.15 12.4 ...
str(summary(medicalcoder_pcccv3.1_current), max.level = 1)
## 'data.frame': 24 obs. of 10 variables:
## $ condition : chr "congeni_genetic" "cvd" "gi" "hemato_immu" ...
## $ label : chr "Other Congenital or Genetic Defect" "Cardiovascular" "Gas
## $ dxpr_or_tech_count : int 8156 171247 56613 37617 85384 210879 15114 39 40205 97720
## $ dxpr_or_tech_percent : num 1.19 24.92 8.24 5.47 12.42 ...
## $ dxpr_only_count : int 8156 131854 41186 37617 85384 209335 896 39 37338 73916 ..
## $ dxpr_only_percent : num 1.19 19.19 5.99 5.47 12.42 ...
## $ tech_only_count : int 0 6159 11327 0 0 244 14210 0 880 4508 ...
## $ tech_only_percent : num 0 0.896 1.648 0 0 ...
## $ dxpr_and_tech_count : int 0 33234 4100 0 0 1300 8 0 1987 19296 ...
## $ dxpr_and_tech_percent: num 0 4.836 0.597 0 0 ...

str(summary(medicalcoder_charlson_cumulative), max.level = 0)
## Warning in .charlson_summary(object): Logic for charlson summary table has been
## implemented for flag.method = 'current'. Using this function for flag.method =
## 'cumulative' may not provide a meaningful summary.
## List of 3
```

The `data.frames` are provided so that end users can easily build their own summary tables using tools such as *knitr* Xie (2015) and *kableExtra* (Zhu 2024)

11 Session Info

Table 8 and Table 9 reports details on the machine, R, and the R packages used to produce this document.

```
si <- sessioninfo::session_info()
si$platform$RAM <- mem_total_gb
si$platform$CPU <- CPU

si$platformDT <-
  data.table::data.table(
    V1 = names(si$platform),
    V2 = do.call(c, si$platform)
  )

si$platformDT <-
  si$platformDT[
    V1 %in% c("version", "os", "system", "CPU", "RAM", "date", "pandoc", "quarto")
  ][c(1:3, 8, 7, 4:6)]
```

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Table 7: Code and resulting table for a summary of the PCCC v2 conditions flagged in the MIMIC-IV dataset.

```
kableExtra::kbl(
  x = summary(medicalcoder_pcccv2.0_current)[, 2:4],
  format = "latex",
  booktabs = TRUE,
  col.names = c("", "Encounters", "Percent"),
  digits = 2
) |>
kableExtra::kable_styling(
  latex_options = c("striped", "scale_down", "HOLD_position")
) |>
kableExtra::pack_rows("Conditions", start_row = 1, end_row = 13) |>
kableExtra::pack_rows("Total Conditions", start_row = 14, end_row = 24)
```

	Encounters	Percent
Conditions		
Other Congenital or Genetic Defect	9960	1.45
Cardiovascular	159359	23.19
Gastrointestinal	56797	8.26
Hematologic or Immunologic	35395	5.15
Malignancy	85183	12.40
Metabolic	211049	30.71
Miscellaneous, Not Elsewhere Classified	17549	2.55
Premature & Neonatal	39	0.01
Neurologic or Neuromuscular	33640	4.90
Renal Urologic	97078	14.13
Respiratory	8272	1.20
Any Technology Dependence	88353	12.86
Any Transplantation	19651	2.86
Total Conditions		
Any Condition	371619	54.08
>= 2 conditions	217327	31.62
>= 3 conditions	92827	13.51
>= 4 conditions	26295	3.83
>= 5 conditions	5382	0.78
>= 6 conditions	786	0.11
>= 7 conditions	73	0.01
>= 8 conditions	12	0.00
>= 9 conditions	0	0.00
>= 10 conditions	0	0.00
>= 11 conditions	0	0.00

Table 8: Details on the machine to generate this document.

Setting	Value
version	R version 4.6.1 (2026-06-24)
os	macOS Tahoe 26.5.1
system	aarch64, darwin23
CPU	Apple M5 Pro
RAM	48.0 GB
date	2026-07-09
pandoc	3.9.0.2 @ /usr/local/bin/ (via rmarkdown)
quarto	1.9.37 @ /usr/local/bin/quarto

Table 9: R packages and versions used to generate this document.

Package Version		Package Version		Package Version		Package Version	
abind	1.4-8	generics	0.1.4	pillar	1.11.1	stringr	1.6.0
backports	1.5.1	ggplot2	4.0.3	pkgconfig	2.0.3	svglite	2.2.2
broom	1.0.13	ggpubr	0.6.3	purrr	1.2.2	systemfonts	1.3.2
car	3.1-5	ggsignif	0.6.4	qwraps2	0.6.3	textshaping	1.0.5
carData	3.0-6	glue	1.8.1	R6	2.6.1	tibble	3.3.1
checkmate	2.3.4	gridExtra	2.3.1	ragg	1.5.2	tidyr	1.3.2
cli	3.6.6	gtable	0.3.6	RColorBrewer	1.1-3	tidyselect	1.2.1
comorbidity	1.1.0	htmltools	0.5.9	Rcpp	1.1.1-1.1	tinytex	0.60
cowplot	1.2.0	jsonlite	2.0.0	rlang	1.3.0	vctrs	0.7.3
data.table	1.18.4	kableExtra	1.4.0	rmarkdown	2.31	viridisLite	0.4.3
digest	0.6.39	knitr	1.51	rstatix	0.7.3	withr	3.0.3
dplyr	1.2.1	labeling	0.4.3	rstudioapi	0.19.0	xfun	0.59
evaluate	1.0.5	lifecycle	1.0.5	S7	0.2.2	xml2	1.6.0
farver	2.1.2	magrittr	2.0.5	scales	1.4.0	yaml	2.3.12
fastmap	1.2.0	medicalcoder	0.8.1.9002	sessioninfo	1.2.4		
Formula	1.2-5	otel	0.2.0	stringi	1.8.7		

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